# Journal Club Preparation

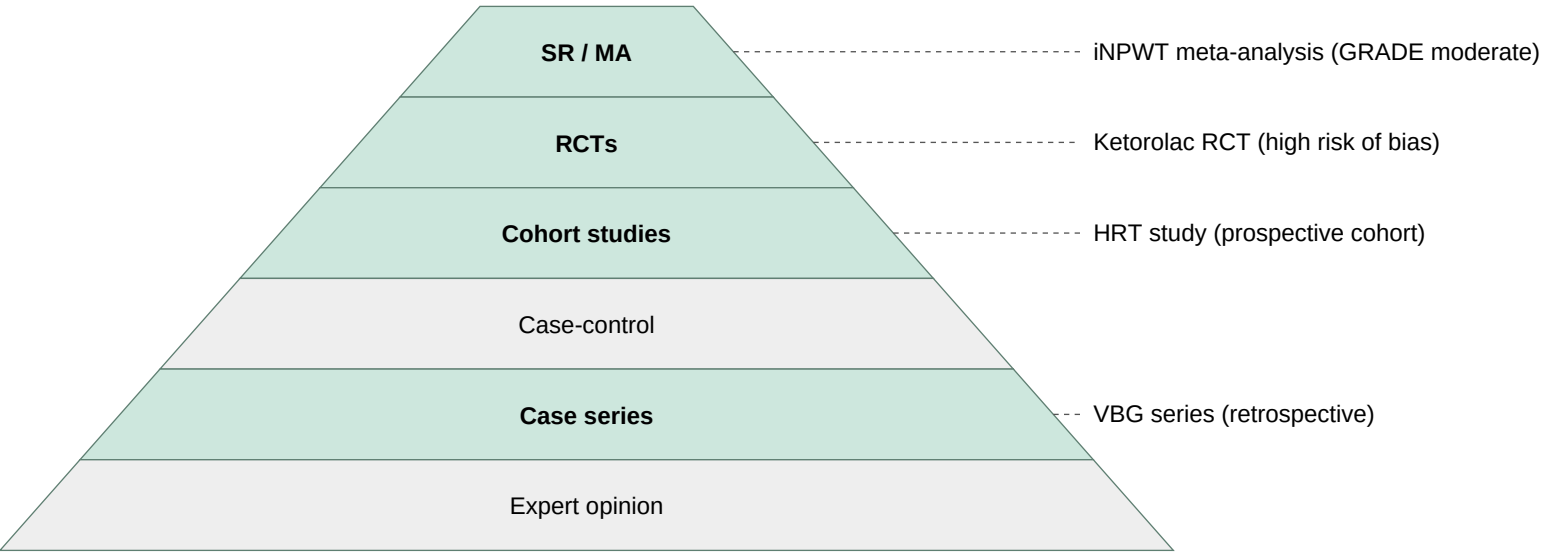
Four articles: critical appraisal, background, limitations and talking points. Prepared 8 November 2026.

Companion file: Journal_Club_Articles.pdf (original articles with every issue highlighted; codes such as K4 or H8 below refer to the tags in that file).

## At a glance

| Article | Design | Sample | Main claim | Evidence level | Biggest problem |
|---|---|---|---|---|---|
| iNPWT for SSI (JAMA Surg 2026) | SR/MA of RCTs | 85 RCTs, 16,980 pts | SSI RR 0.64 (NNT 21); blistering NNH 18 | Top tier; GRADE moderate | Pooling very different surgeries; harms vs benefits |
| Ketorolac in breast surgery (PRS GO 2025) | 3-arm RCT | 63 (or 66, 73, 65) | 30 mg reduces pain and opioid use | RCT tier, high risk of bias | Numbers do not add up; trivial effect sizes |
| HRT before brain death (JTACS 2026) | Prospective observational cohort | 1,184 pts, 38 centers | HRT raises donation (aOR 4.60); MP + levo best | Cohort tier | Confounding, immortal time, reverse causation |
| Spinoplastic VBG (JPRAS 2026) | Retrospective series + controls | 18 vs 18 pts | VBG safe, no extra OR time or bleeding | Case series tier | Fusion never measured; no power |

## Where each study sits on the evidence pyramid




The pyramid ranks study **designs**, not the quality of a given paper. GRADE starts from the design and then rates down for risk of bias, inconsistency, imprecision, indirectness and publication bias. The ketorolac RCT shows why: high on the pyramid, low in reliability. The VBG paper has a control group but is honestly a level IV feasibility report; the HRT study is labeled Level II by the journal, but as a therapy question it is a cohort with strong confounding (Oxford 2b to 3).

**Common thread for the discussion:** three of the four papers either treat a non-significant result as proof of safety or equivalence (ketorolac hematoma, VBG vs control) or draw causal and ranking conclusions the design cannot support (HRT combinations, ketorolac doses). The iNPWT meta-analysis is the only one with solid evidence of efficacy; the useful debate there is the balance of benefit and harm by baseline risk, not whether the effect exists.

JAMA Surgery | Original Investigation

# Incisional Negative Pressure Wound Therapy for Prevention of Surgical Site Infection
## A Systematic Review and Meta-Analysis

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Invited Commentary page 1032

Supplemental content

**IMPORTANCE** Incisional negative pressure wound therapy (iNPWT) has been proposed as a nonantibiotic adjunct to standard dressings for preventing surgical site infection (SSI), but uncertainty remains regarding consistency across SSI subtypes and dressing-related harms.

**OBJECTIVE** To evaluate the efficacy and safety of iNPWT vs standard dressing in adults undergoing surgery.

**DATA SOURCES** PubMed, Embase, and Cochrane Central Register of Controlled Trials (CENTRAL) were searched from inception through March 7, 2026.

**STUDY SELECTION** Randomized clinical trials (RCTs) enrolling adults in whom iNPWT was applied to primarily closed surgical incisions and compared with standard dressing.

**DATA EXTRACTION AND SYNTHESIS** Two reviewers independently extracted data. Risk ratios (RRs) and mean differences with 95% CIs were pooled using random-effects models.

**MAIN OUTCOMES AND MEASURES** The primary outcome was overall SSI. Secondary outcomes included SSI subtypes, wound dehiscence, seroma, skin necrosis, reoperation, readmission, mortality, skin blistering, and device-related adverse events.

**RESULTS** Eighty-five RCTs comprising 16 980 patients (iNPWT, n = 8463; standard dressing, n = 8517) were included. The mean (SD) age was 48.3 (19.4) years (48.1 [19.5] years in the iNPWT group and 48.5 [19.4] years in the standard dressing group), and 5660 of 16 075 individuals (35.2%) were male (2889 [35.7%] in the iNPWT group and 2771 [34.7%] in the standard dressing group). iNPWT was associated with lower risk of overall SSI (RR, 0.64; 95% CI, 0.57-0.72), deep SSI (RR, 0.66; 95% CI, 0.51-0.86), superficial SSI (RR, 0.59; 95% CI, 0.49-0.72), wound dehiscence (RR, 0.73; 95% CI, 0.61-0.88), seroma (RR, 0.77; 95% CI, 0.63-0.94), reoperation (RR, 0.80; 95% CI, 0.64-0.99), and skin necrosis (RR, 0.38; 95% CI, 0.18-0.80). No significant differences were observed for organ-space SSI, readmission, or mortality. iNPWT was associated with higher risks of skin blistering (RR, 4.51; 95% CI, 2.37-8.58) and device-related adverse events (RR, 11.87; 95% CI, 4.74-29.74). The number needed to treat was 21 for SSI; the number needed to harm was 18 for skin blistering and 11 for device-related adverse events. Trial sequential analysis confirmed evidence for overall, deep, and superficial SSI. Certainty of evidence was moderate.

**CONCLUSIONS AND RELEVANCE** In this systematic review and meta-analysis of 85 RCTs, iNPWT was associated with lower risks of overall, deep, and superficial SSI and several wound complications but potentially higher risks of skin blistering and device-related adverse events. These findings support selective rather than routine use, particularly in patients at elevated risk of incisional wound complications.

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Surgical site infection (SSI) is a leading cause of postoperative morbidity.[1] Despite contemporary perioperative prevention bundles, SSI continues to drive prolonged hospitalization, reoperation, and excess health care costs across surgical specialties.[2,3] Further reductions in infection rates have proven difficult to achieve consistently, particularly in higher-risk operative populations, and SSI prevention remains a priority within surgical quality improvement.[4]

Incisional negative pressure wound therapy (iNPWT), applied to primary closed surgical incisions, has been proposed as a nonantibiotic adjunct to standard dressings. The device removes exudate, reduces edema and lateral incisional tension, and protects the local wound environment.[5] Randomized clinical trials (RCTs) across lower-extremity fracture surgery, cesarean delivery, vascular surgery, and abdominal wall reconstruction have reported lower rates of postoperative wound complications with iNPWT in selected higher-risk patients.[6-9] Within contemporary antimicrobial stewardship frameworks, a nonantibiotic approach to SSI reduction is particularly appealing.[10,11]

Prior meta-analyses have generally favored iNPWT for overall SSI but have been limited by heterogeneity in surgical populations, wound class, device platform, negative-pressure setting, intended duration of therapy, comparator dressings, follow-up intervals, and outcome definitions.[12-16] Whether the reported benefit extends to deep and organ-space SSI and whether it is offset by dressing-related harms, such as skin blistering and device-related adverse events, have not been clearly established.[17]

We therefore conducted an updated systematic review and meta-analysis restricted to RCTs comparing iNPWT with standard dressing in adults undergoing surgery. We evaluated overall SSI as the primary outcome and assessed deep, superficial, and organ-space infection, wound complications, adverse events, and hospital length of stay as secondary outcomes, with prespecified analyses to explore whether treatment effects differed across device, wound, and procedural characteristics.

### Key Points

**Question** Is incisional negative pressure wound therapy (iNPWT) associated with reduced surgical site infection (SSI) and wound complications compared with standard dressing?

**Findings** In this systematic review and meta-analysis of 85 randomized clinical trials including 16 980 patients, iNPWT was associated with lower risks of overall, deep, and superficial SSI and several wound complications but with higher risks of skin blistering and device-related adverse events. The number needed to treat for SSI was 21; the number needed to harm for skin blistering was 18.

**Meaning** These findings support selective use of iNPWT in patients at elevated risk of incisional wound complications rather than routine application across all surgical settings.

## Methods

### Protocol and Registration

This systematic review and meta-analysis was conducted and reported in accordance with the *Cochrane Handbook for Systematic Reviews of Interventions*[18] and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting guidelines.[19] The study protocol was prospectively registered on PROSPERO (PROSPERO identifier: CRD420261367529). As this study used only previously published, aggregate-level data, institutional review board approval and informed consent were not required.

### Search Strategy and Screening

PubMed, Embase, and Cochrane Central Register of Controlled Trials (CENTRAL) were searched from inception to March 7, 2026. The search combined controlled vocabulary and free-text terms related to iNPWT and SSI (eTable 1 in Supplement 1). Reference lists of included articles and prior review articles were additionally screened. Two reviewers independently performed title and abstract screening, followed by full-text eligibility assessment. Disagreements were resolved through consensus, with senior adjudication when required. Interrater reliability between reviewers was excellent (Cohen $\kappa = 0.94$).

### Inclusion Criteria

We included RCTs enrolling adults (aged ≥16 years) undergoing operative procedures in which iNPWT was applied to primary closed surgical incisions and compared with standard wound dressing. Studies were required to report at least 1 predefined efficacy or safety outcome. We excluded nonrandomized studies, single-arm studies, trial registrations without reported results, review articles, conference abstracts without full data, and nonhuman studies.

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### Data Extraction

Data were collected independently by 2 reviewers using a predefined extraction template. Information obtained from each study included trial characteristics (authors, year, country, enrollment period, registry information), sample size, surgical setting, wound class, iNPWT platform, negative-pressure setting, intended treatment duration, comparator dressing, follow-up duration, outcome definitions, and reported efficacy and safety outcomes. Discrepancies were resolved through consensus.

### Outcomes and Subgroup Analyses

The primary outcome was overall SSI. Secondary outcomes included deep SSI, superficial SSI, organ-space infection, wound dehiscence, seroma, fluid leak, skin necrosis, hematoma, cellulitis, wound bleeding, pain, reoperation, readmission, mortality, serious and device-related adverse events (defined in eTable 2 in Supplement 1), skin blistering or reaction, and hospital length of stay.

Prespecified subgroup analyses were performed, where data permitted, according to device pressure, intended duration of therapy, dressing type, wound class, and surgical specialty (abdominal, vascular, orthopedic or trauma, plastic, obstetric, breast, and cardiac surgery).



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### Risk of Bias and Certainty of Evidence

Two reviewers independently assessed risk of bias using the Cochrane Risk of Bias 2 (RoB 2) tool[20] and certainty of evidence using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework.[21] Disagreements were resolved through consensus with senior adjudication when required. Publication bias was assessed through visual inspection of funnel plots and Egger regression testing when 10 or more studies were available.

### Statistical Analysis

All statistical analyses were performed using R version 4.4.2 statistical software (R Foundation for Statistical Computing).[22] Random-effects meta-analyses were conducted using restricted maximum likelihood estimation to account for anticipated clinical and methodological heterogeneity. Dichotomous outcomes were summarized as risk ratios (RRs) and continuous outcomes as mean differences, each with 95% CIs. Statistical heterogeneity was assessed using Cochran *Q* test and quantified using the $I^2$ statistic (>25% suggesting meaningful heterogeneity).

Robustness of statistically significant findings was examined using leave-1-out analyses and sensitivity analyses excluding trials judged at high risk of bias. Study influence and contribution to heterogeneity were further evaluated using Baujat plots. Trial sequential analysis was conducted for selected outcomes (2-sided α = .05; 90% power; required information size based on the observed intervention effect), comparing the cumulative *z* curve against conventional (±1.96) and O'Brien-Fleming-type boundaries. Absolute risk differences were calculated for selected outcomes, from which the number needed to treat (NNT) for benefit and the number needed to harm (NNH) were derived.

### Critical Appraisal of Prior Systematic Reviews

Prior reviews of iNPWT were assessed with AMSTAR 2[23] by 2 reviewers.

## Results

### Study Selection and Characteristics

The database search identified 1261 records (CENTRAL, 288; Embase, 446; PubMed, 527). After removal of duplicates and screening, 85 RCTs comprising 16 980 patients (iNPWT, n = 8463; standard dressing, n = 8517) were included (**Figure 1**).

Baseline characteristics are summarized in the **Table**. The mean (SD) age was 48.3 (19.4) years (48.1 [19.5] years in the iNPWT group and 48.5 [19.4] years in the standard dressing group), and 5660 of 16 075 individuals (35.2%) were male (2889 [35.7%] in the iNPWT group and 2771 [34.7%] in the standard dressing group). The mean (SD) body mass index (calculated as weight in kilograms divided by height in meters squared) was approximately 31 in both groups (31.0 [8.6] in the iNPWT group and 30.6 [8.3] in the standard dressing group), and diabetes was present in approximately 16%. The most common surgical settings were obstetric and gynecologic (33.5%), abdominal (29.9%), and trauma (12.9%). Wound class was predominantly clean (40.9%) or clean contaminated (52.5%). Comparator dressings included gauze, hydrocolloid dressings, and silicone gauze. iNPWT devices included PICO (4325 patients; Smith + Nephew), PREVENA (3000 patients; 3M), and other platforms. Detailed trial characteristics, including SSI definitions, are provided in eTables 2 through 12 in Supplement 1.

### Primary Outcome: Overall SSI

Overall SSI was reported in 82 trials (96.5%). iNPWT was associated with a lower risk of overall SSI compared with standard dressing (800 of 8338 participants [9.6%] vs 1207 of 8407 participants [14.4%]; RR, 0.64; 95% CI, 0.57-0.72; $P < .001$; $I^2 = 31.6\%$) (eFigure 1 in Supplement 1).

### Secondary Outcomes

#### SSI Subtypes

iNPWT was associated with lower risk of deep SSI (RR, 0.66; 95% CI, 0.51-0.86; $P = .002$; $I^2 = 0.0\%$) and superficial SSI (RR, 0.59; 95% CI, 0.49-0.72; $P < .001$; $I^2 = 22.5\%$). No significant difference was observed for organ-space infection (eFigures 2-4 in Supplement 1).

#### Wound Complications and Clinical Outcomes

iNPWT was associated with lower risk of wound dehiscence (RR, 0.73; 95% CI, 0.61-0.88; $P < .001$; $I^2 = 10.4\%$), seroma (RR, 0.77; 95% CI, 0.63-0.94; $P = .01$; $I^2 = 4.8\%$), fluid leak (RR, 0.70; 95% CI, 0.58-0.85; $P < .001$; $I^2 = 20.1\%$), skin necrosis (RR, 0.38; 95% CI, 0.18-0.80; $P = .01$; $I^2 = 0.0\%$), reoperation (RR, 0.80; 95% CI, 0.64-0.99; $P = .04$; $I^2 = 0.0\%$), and hospital length of stay (mean difference, −0.67 days; 95% CI, −1.18 to −0.16; $P = .01$; $I^2 = 84.9\%$) (eFigures 5-10 in Supplement 1). No significant differences were observed for readmission, hematoma, cellulitis, wound bleeding, pain, serious adverse events, and mortality (eFigures 11-17 in Supplement 1).

#### Adverse Events

iNPWT was associated with a substantially higher risk of skin blistering or reaction (RR, 4.51; 95% CI, 2.37-8.58; $P < .001$; $I^2 = 46.8\%$) and device-related adverse events (RR, 11.87; 95% CI, 4.74-29.74; $P < .001$; $I^2 = 0.0\%$) (eFigures 18 and 19 in Supplement 1). Effect sizes for secondary outcomes are summarized in **Figure 2**.

### Absolute Treatment Effects

Absolute risk differences corresponded to NNTs of 21 to prevent 1 overall SSI, 74 to prevent 1 deep SSI, and 25 to prevent 1 superficial SSI. The corresponding NNH was 18 for skin blistering or reaction and 11 for device-related adverse events.

### Sensitivity and Influence Analyses

Leave-1-out analyses demonstrated directional consistency of pooled estimates for overall SSI (RR range, 0.63-0.65), deep SSI (RR range, 0.61-0.73), superficial SSI (RR range, 0.57-0.61), wound dehiscence, seroma, and hospital length of stay, all of which remained statistically significant after sequential exclusion of individual trials (eFigures 20-25 in Supplement 1). For fluid leak, omission of the study by Costa et al[9] or the study by Hyldig et al[24] attenuated the pooled estimate across the null



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(eFigure 26 in Supplement 1). For skin necrosis, omission of the study by Witt-Majchrzak et al[25] attenuated the estimate to the margin of significance (eFigure 27 in Supplement 1). The excess risks of skin blistering and device-related adverse events remained directionally consistent across all study exclusions (eFigures 28 and 29 in Supplement 1).

Sensitivity analyses excluding 5 trials judged at high risk of bias[26-30] yielded results consistent with the primary analysis for overall SSI (RR, 0.64; 95% CI, 0.57-0.72), deep SSI (RR, 0.67; 95% CI, 0.53-0.84), superficial SSI (RR, 0.59; 95% CI, 0.48-0.71), and wound dehiscence (RR, 0.74; 95% CI, 0.63-0.87) (eTable 13 in Supplement 1).

Baujat plots identified a small number of influential trials for overall, deep, and superficial SSI, without changing the overall interpretation (eFigures 30-32 in Supplement 1).

## Subgroup Analyses

Significant interaction tests were observed for overall SSI by dressing type and wound class. Similarly, significant interactions were observed for superficial SSI by wound class and surgical specialty, wound dehiscence by wound class and surgical specialty, hospital length of stay by device pressure and wound class, and readmission by device pressure.

No significant interaction was observed for overall SSI by device pressure, intended duration of therapy, or surgical specialty. Deep SSI, organ-space infection, seroma, hematoma, reoperation, and skin blistering showed no significant effect modification across the examined subgrouping variables (eFigures 33-84 in Supplement 1).

## Trial Sequential Analyses

Trial sequential analysis demonstrated firm cumulative evidence for overall SSI (required information size [RIS], 653; cumulative $z = -7.31$), deep SSI (RIS, 3588; cumulative $z = -3.17$), superficial SSI (RIS, 862; cumulative $z = -6.18$), wound dehiscence (RIS, 1444; cumulative $z = -2.95$), fluid leak (RIS 437; cumulative $z = -3.75$), seroma (RIS, 3082; cumulative $z = -2.64$), skin necrosis (RIS, 1333; cumulative $z = -2.44$), length of stay (RIS, 4738; cumulative $z = -3.98$), and skin blistering or reaction (RIS, 221; cumulative $z = 6.69$) (eFigures 85-93 in Supplement 1).

Evidence remained inconclusive for wound bleeding (RIS, 114 847; cumulative $z = -0.57$), cellulitis (RIS, 12 173; cumulative $z = -0.92$), pain (RIS, 2332; cumulative $z = -0.98$), and hematoma (RIS, 6737; cumulative $z = -1.89$), indicating that additional data would be required to exclude clinically meaningful effects for these outcomes (eFigures 94-97 in Supplement 1).

## Publication Bias

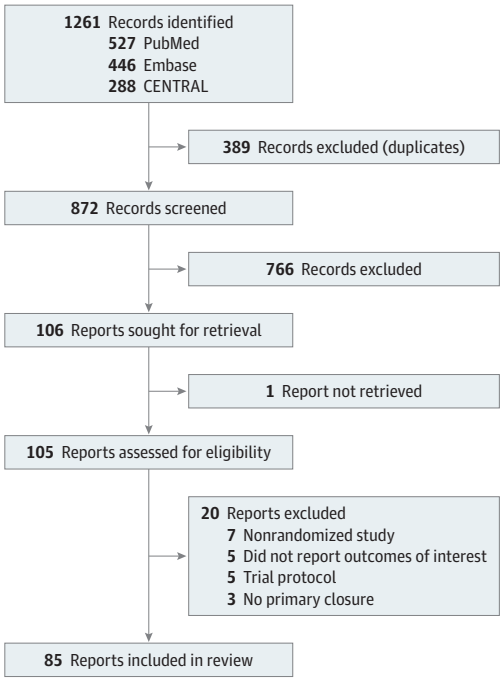
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Funnel plots suggested asymmetry for overall and superficial SSI, whereas the plot for deep SSI appeared more symmetric. Egger regression tests did not demonstrate significant small-study effects for overall, deep, or superficial SSI (eFigure 98 in Supplement 1).

## Risk of Bias and Certainty of Evidence

Among the 85 included trials, 5 were judged as having high risk of bias, 71 as having some concerns, and 9 as having low risk of bias according to Cochrane RoB 2 (eTable 14 in Supplement 1). Certainty of evidence for the primary outcomes was moderate, with downgrading primarily attributable to risk of bias (eTable 15 in Supplement 1).

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Flow Diagram of Study Identification, Screening, and Inclusion




The flow diagram shows identification, screening, eligibility assessment, and inclusion of randomized clinical trials evaluating incisional negative pressure wound therapy compared with standard dressing.
CENTRAL indicates Cochrane Central Register of Controlled Trials.

## Critical Appraisal of Prior Systematic Reviews

AMSTAR 2 appraisal of 6 prior iNPWT meta-analyses found 1 high-confidence review,[17] 2 moderate,[13,31] 1 low,[14] and 2 critically low[15,32] (eTable 16 in Supplement 1). The present review had high confidence, and its overall SSI estimate was compared with prior meta-analyses (**Figure 3**).

# Discussion

In this meta-analysis of 85 RCTs including 16 980 patients, iNPWT applied to primary closed surgical incisions was associated with lower risks of overall, deep, and superficial SSI, wound dehiscence, seroma, fluid leak, skin necrosis, reoperation, and shorter hospital length of stay compared with standard dressing. No differences were observed for organ-space infection, readmission, mortality, cellulitis, wound bleeding, pain, or serious adverse events. These benefits were offset by potentially higher risks of skin blistering or reaction and device-related adverse events. Absolute treatment effects indicate 1 SSI prevented per 21 treated and 1 dressing-related adverse event per 11 treated.



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Table. Baseline Patient and Operative Characteristics Among 85 Randomized Clinical Trials

| Characteristic | iNPWT | Standard dressing |
|---|---|---|
| Patients, No. | 8463 | 8517 |
| iNPWT device type, No. | | |
| PICO | 4325 | NA |
| PREVENA | 3000 | |
| VAC | 412 | |
| CuraVAC | 80 | |
| MJ-01B3 | 50 | |
| VivanoTec | 46 | |
| NANOVA | 35 | |
| VSD | 33 | |
| Avelle | 30 | |
| RENASYS TOUCH | 30 | |
| CCNPWT | 25 | |
| NR | 397 | |
| iNPWT pressure, mm Hg, No. | | |
| 125 | 3607 | NA |
| 120 | 66 | |
| 100 | 70 | |
| 90 | 45 | |
| 80 | 4289 | |
| 75 | 55 | |
| Cyclical 50-125 | 50 | |
| Cyclical 50-200 | 20 | |
| NR | 261 | |
| Age, mean (SD), y | 48.1 (19.5) | 48.5 (19.4) |
| Sex, No. (%) | (n = 8085) | (n = 7990) |
| Female | 5196 (64.3) | 5219 (65.3) |
| Male | 2889 (35.7) | 2771 (34.7) |
| BMI, mean (SD) | 31.0 (8.6) | 30.6 (8.3) |
| Diabetes, No./No. (%) | 1228/7680 (16.0) | 1160/7541 (15.4) |
| Smoker, No./No. (%) | 1460/7656 (19.1) | 1589/7516 (21.1) |
| Albumin, mean (SD), g/dL | 3.72 (0.84) | 3.74 (0.82) |
| Immunosuppressed, No./No. (%) | 113/1675 (6.7) | 106/1690 (6.3) |
| ASA score, mean (SD) | 2.49 (0.76) | 2.50 (0.76) |
| Surgical specialty, No. (%)[a] | | |
| Obstetrics and gynecology | 2842 (32.3) | 2846 (32.1) |
| Abdominal | 2555 (29.0) | 2518 (28.4) |
| Trauma | 1085 (12.3) | 1099 (12.4) |
| Orthopedic | 747 (8.5) | 771 (8.7) |
| Vascular | 629 (7.1) | 695 (7.8) |
| Breast | 234 (2.7) | 238 (2.7) |
| Cardiac | 195 (2.2) | 195 (2.2) |
| Plastic | 122 (1.4) | 111 (1.3) |
| General | 54 (0.6) | 44 (0.5) |
| Elective vs emergency, No. (%) | | |
| Elective | 3163 (56.9) | 3206 (57.4) |
| Emergency | 2391 (43.1) | 2380 (42.6) |
| Wound class, No. (%)[b] | | |
| I | 3195 (40.6) | 3247 (41.2) |
| II | 4120 (52.4) | 4147 (52.6) |
| III | 353 (4.5) | 300 (3.8) |
| IV | 196 (2.5) | 183 (2.3) |

(continued)

Table. Baseline Patient and Operative Characteristics Among 85 Randomized Clinical Trials (continued)

| Characteristic | iNPWT | Standard dressing |
|---|---|---|
| Operative time, mean (SD), min | 117.2 (103.9) | 117.9 (108.0) |
| Blood loss, mean (SD), mL | 546.0 (515.0) | 540.1 (519.3) |
| Incision length, mean (SD), cm | 20.3 (9.9) | 20.0 (10.0) |

Abbreviations: ASA, American Society of Anesthesiologists; BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); CCNPWT, continuous controlled negative pressure wound therapy; iNPWT, incisional negative pressure wound therapy; NA, not applicable; NR, not reported; VAC, vacuum-assisted closure; VSD, vacuum sealing drainage.

SI conversion factor: To convert albumin to grams per liter, multiply by 10.

[a] Surgical specialty percentages are calculated from patients with available data.

[b] Wound class is defined according to the Centers for Disease Control and Prevention classification: I indicates clean; II, clean contaminated; III, contaminated; IV, dirty.

These results are broadly consistent with prior meta-analyses[12-17] and extend that evidence in several respects: this is the largest analysis to date to our knowledge, incorporates trials published through 2026, and provides a comprehensive assessment of both SSI subtypes and dressing-related adverse events. Groenen et al[17] concluded that cumulative evidence for the primary outcome had crossed trial sequential analysis monitoring boundaries, consistent with our own trial sequential analysis. Further RCTs powered for overall SSI are therefore unlikely to be informative. However, several important questions remain unresolved, and our analysis highlights areas in which interpretation of the aggregate evidence requires caution.

First, the overall pooled benefit must be reconciled with the results of large, well-conducted trials that individually report null findings. The WHIST (Wound Healing in Surgery for Trauma) trial,[9] the largest single RCT of iNPWT in lower-extremity fracture surgery (1548 patients), found no reduction in deep SSI. The SUNRRISE (Single Use Negative Pressure Dressing for Reduction in Surgical Site Infection Following Emergency Laparotomy) trial,[33] a pragmatic, assessor-masked trial of 840 patients undergoing emergency laparotomy, found no difference in SSI (28.4% among those receiving iNPWT vs 27.4% in the surgeon's preference group; RR, 1.03; 95% CI, 0.83-1.28). These results do not necessarily contradict the present meta-analysis, but they do indicate that the benefit of iNPWT is not uniform across surgical contexts. In the present analysis, funnel plots suggested asymmetry for overall and superficial SSI, raising the possibility of small-study effects. Although Egger test results did not reach conventional significance, the pattern is consistent with the observation that many of the largest and most methodologically rigorous individual trials have reported attenuated or null effects, whereas smaller single-center studies have tended to report larger treatment effects. This does not invalidate the pooled estimate, but it warrants caution regarding its magnitude.

Second, the pattern of benefit across SSI subtypes provides mechanistic insight. iNPWT was associated with reductions in superficial and deep SSI but not organ-space infection. This dissociation is consistent with a localized mechanism

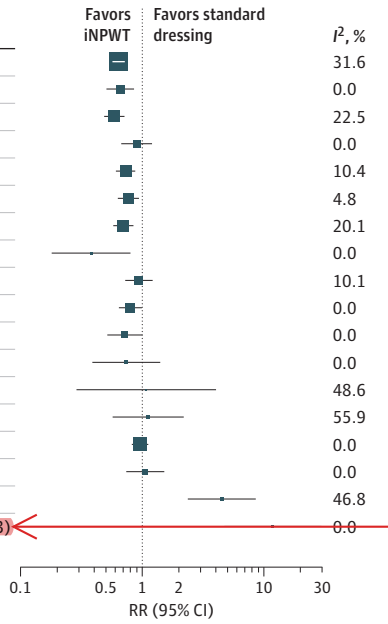


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Figure 2. Summary Forest Plot for All Efficacy and Safety Outcomes

| Outcome | iNPWT No. of events | iNPWT Total No. | Standard dressing No. of events | Standard dressing Total No. | RR (95% CI) | $I^2$, % |
|---|---|---|---|---|---|---|
| Overall SSI | 800 | 8338 | 1207 | 8407 | 0.64 (0.57-0.72) | 31.6 |
| Deep SSI | 127 | 5376 | 202 | 5420 | 0.66 (0.51-0.86) | 0.0 |
| Superficial SSI | 283 | 4533 | 476 | 4613 | 0.59 (0.49-0.72) | 22.5 |
| Organ-space SSI | 67 | 3665 | 83 | 3694 | 0.90 (0.67-1.20) | 0.0 |
| Wound dehiscence | 384 | 5430 | 499 | 5475 | 0.73 (0.61-0.88) | 10.4 |
| Seroma | 177 | 3752 | 245 | 3781 | 0.77 (0.63-0.94) | 4.8 |
| Fluid leak | 145 | 1335 | 211 | 1319 | 0.70 (0.58-0.85) | 20.1 |
| Skin necrosis | 7 | 630 | 36 | 657 | 0.38 (0.18-0.80) | 0.0 |
| Readmission | 150 | 3846 | 162 | 3890 | 0.94 (0.73-1.22) | 10.1 |
| Reoperation | 132 | 3950 | 179 | 4038 | 0.80 (0.64-0.99) | 0.0 |
| Hematoma | 55 | 3695 | 90 | 3718 | 0.72 (0.52-1.00) | 0.0 |
| Cellulitis | 16 | 1107 | 22 | 1101 | 0.74 (0.39-1.41) | 0.0 |
| Wound bleeding | 25 | 1963 | 21 | 1955 | 1.08 (0.29-4.05) | 48.6 |
| Pain | 194 | 1823 | 200 | 1792 | 1.12 (0.57-2.20) | 55.9 |
| SAE | 221 | 2705 | 222 | 2728 | 0.96 (0.82-1.12) | 0.0 |
| Mortality | 59 | 3887 | 55 | 3917 | 1.06 (0.74-1.52) | 0.0 |
| Skin blistering or reaction | 227 | 3285 | 44 | 3292 | 4.51 (2.37-8.58) | 46.8 |
| Device-related adverse events | 69 | 745 | 0 | 823 | 11.87 (4.74-29.73) | 0.0 |

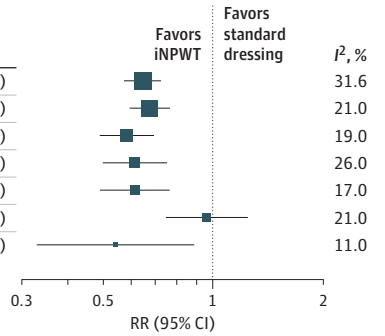



J7

Summary of pooled effect estimates for all prespecified efficacy and safety outcomes comparing incisional negative pressure wound therapy (iNPWT) with standard dressing. The size of each square is proportional to the total sample size. $I^2$ values denote the proportion of total variability attributable to between-study heterogeneity. RR indicates risk ratio; SAE, serious adverse event; SSI, surgical site infection.

Figure 3. Forest Plot Comparing Cumulative Meta-Analytic Evidence for Overall Surgical Site Infection

| Meta-analysis | Trials, No. | GRADE rating | iNPWT No. of events | iNPWT Total No. | Standard dressing No. of events | Standard dressing Total No. | RR (95% CI) | $I^2$, % |
|---|---|---|---|---|---|---|---|---|
| Current meta-analysis | 82 | Moderate | 800 | 8338 | 1207 | 8407 | 0.64 (0.57-0.72) | 31.6 |
| Groenen et al,[17] 2023 | 57 | High | 540 | 6849 | 802 | 6985 | 0.67 (0.59-0.76) | 21.0 |
| Li et al,[31] 2019 | 45 | Low | 280 | 3285 | 474 | 3339 | 0.58 (0.49-0.69) | 19.0 |
| Shiroky et al,[13] 2020 | 32 | Moderate | 229 | 2354 | 373 | 2350 | 0.61 (0.50-0.75) | 26.0 |
| Zwanenburg et al,[14] 2020 | 28 | High | 194 | 2193 | 315 | 2205 | 0.61 (0.49-0.76) | 17.0 |
| Ge,[32] 2018 | 17 | Not reported | 91 | 928 | 97 | 930 | 0.96 (0.74-1.24) | 21.0 |
| Hyldig et al,[15] 2016 | 7 | Moderate | 30 | 634 | 55 | 617 | 0.54 (0.33-0.89) | 11.0 |




Comparison of pooled effect estimates for overall surgical site infection across successive meta-analyses evaluating incisional negative pressure wound therapy (iNPWT) vs standard dressing. The size of each square is proportional to the total sample size. Grading of Recommendations Assessment, Development, and Evaluation (GRADE) certainty-of-evidence ratings are shown for each meta-analysis. RR indicates risk ratio.

of action at the incision site—removal of exudate, reduction of edema and lateral tension, improved wound-edge apposition, and protection of the local wound environment[5]—rather than a broader systemic antibacterial effect. This interpretation also offers a plausible explanation for the null results observed in emergency laparotomy populations, where SSI pathogenesis is predominantly driven by intra-abdominal contamination rather than incisional factors.[33-35] By contrast, benefit may be greatest in clean or clean-contaminated procedures where the principal risk relates to incisional wound failure rather than peritoneal microbial exposure.[36,37] The significant subgroup interaction by wound class observed in this analysis is consistent with this hypothesis, although trial-level subgroup comparisons should be interpreted as exploratory.

Third, the adverse event profile of iNPWT has been incompletely characterized in prior syntheses. Groenen et al[17] reported a 5-fold increase in skin blistering but did not report device-related adverse events as a separate outcome. The present analysis quantifies both: skin blistering or reaction (RR, 4.51; 95% CI, 2.37-8.58) and device-related adverse events (RR, 11.87; 95% CI, 4.74-29.74), albeit based on a limited number of trials with wide 95% CIs. The absolute risk data are particularly informative: the NNT of 21 to prevent 1 SSI is essentially

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equivalent to the NNH of 18 for skin blistering. Although skin blistering is generally regarded as a minor and self-limiting complication,[17] this near equivalence between benefit and harm reinforces the need for risk-stratified use. In low-risk settings, the net clinical benefit may be marginal or absent.

A formal cost-effectiveness analysis was beyond the scope of this synthesis as few included trials reported economic outcomes (eTable 12 in Supplement 1). Nonetheless, absolute treatment effects permit an illustrative threshold to contextualize clinical findings. With an NNT of 21 for overall SSI and a single-use device priced at $150 to $500 USD (depending on platform and market), iNPWT becomes approximately cost neutral when the cost attributable to an SSI exceeds $3150 to $10 500. Reported mean costs for SSI commonly exceed this range,[38-40] particularly for deep or organ-space infection and within orthopedic, cardiac, and vascular surgery,[41-44] with economic benefit often concentrated in higher-risk populations.[42,45] As such, decisions to adopt iNPWT should be guided by baseline SSI risk and local cost structures rather than by efficacy alone.

These observations align with current guidance. The World Health Organization[46] conditionally recommends prophylactic iNPWT on primarily closed high-risk incisions, and both the American College of Surgeons/Surgical Infection Society[10] and American Society of Colon and Rectal Surgeons[47] have issued comparable conditional recommendations, noting that benefit is most pronounced where baseline SSI risk is elevated.

Future research should shift from broad efficacy comparisons toward patient-level identification of those most likely to benefit. The present analysis, like its predecessors, is a trial-level synthesis that cannot resolve which individual patient characteristics, device protocols, or procedural contexts confer the greatest net benefit. Individual participant data meta-analysis, stratified by wound class, procedural risk, and device parameters, would provide more actionable evidence for clinical practice and guideline development.[48,49] Standardization of adverse event reporting across iNPWT trials will enable precise characterization of the benefit-harm balance. Importantly, these data will be needed for targeted cost-effectiveness analyses to define exact thresholds for selective deployment of iNPWT in high-risk subgroups.

### Limitations

Several limitations should be considered. First, the included trials spanned diverse surgical procedures, wound classes, device platforms, pressure settings, comparator dressings, and follow-up intervals. Although this breadth improves generalizability, it introduces clinical heterogeneity that cannot be fully resolved in a trial-level analysis. Second, outcome definitions were not standardized across trials. In particular, Centers for Disease Control and Prevention–based SSI definitions may not align with contemporary fracture-related infection frameworks[50,51] or periprosthetic joint infection criteria,[52,53] potentially introducing outcome misclassification in orthopedic and trauma studies. Third, blinding of a visible device intervention is inherently difficult, and performance bias cannot be excluded despite efforts at outcome assessor blinding. Fourth, comparator dressings and cointerventions were variably reported, and adverse event reporting was inconsistent across trials. Fifth, the high heterogeneity observed for hospital length of stay ($I^2$ = 84.9%) limits confidence in the precision of that estimate. Finally, subgroup analyses were based on study-level data and should be considered exploratory.

## Conclusions

In this systematic review and meta-analysis of 85 RCTs, iNPWT was associated with lower risks of overall, deep, and superficial SSI and several incision-related wound complications but potentially higher risks of skin blistering or reaction and device-related adverse events. Cumulative evidence for overall SSI benefit is now saturated, indicating that further trials for this outcome are unlikely to be informative. Future cost-effectiveness analyses are necessary to define precise thresholds for iNPWT deployment.


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Invited Commentary

# Considering Incisional Negative-Pressure Wound Therapy

Marja A. Boermeester, MD, PhD

**Patel et al**[1] performed a meta-analysis of 85 randomized clinical trials (RCTs) comprising 16 980 adult patients in whom prophylactic incisional negative pressure wound therapy (piNPWT) was applied to primarily closed surgical incisions and compared with standard dressings. piNPWT was associated with lower risk of overall surgical site infection (SSI) (risk ratio [RR], 0.64; 95% CI, 0.57-0.72; $I^2$ = 31.6%). Their meta-analysis provides exhaustive supplementary data and granular subgroup analyses, such as level of negative pressure, surgery type, surgical site infection type, other surgical site outcomes, and NPWT device. Most data come from 1 of 2 commercially available devices: the PICO system (Smith & Nephew) in 4325 patients and PREVENA (3M Solventum) in 3000 patients, while the rest of the trials used miscellaneous devices.

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This message, however, is not new. Patel et al[1] performed comprehensive updated syntheses that strengthen and add nuance to the evidence base for incisional negative-pressure wound therapy by adding 28 randomized clinical trials to prior meta-analyses, among which the 2023 meta-analysis by Groenen et al[2] has been frequently cited and follows the same principles. That meta-analysis yielded an RR of 0.67 (95% CI, 0.59-0.76; $I^2$ = 21%) for piNPWT compared with standard dressing in 57 RCTs. Compared with previous meta-analyses, the relative risk stabilized, and the confidence interval narrowed. In the 2023 trial sequential analysis, the cumulative $z$ curve crossed the trial sequential monitoring boundary for benefit, confirming the robustness of the summary effect estimate from the aggregated trial results.

Based on their data, Patel et al advise "selective use of piNPWT" based on a number needed to treat of 21, but that conclusion is debatable, given the high median costs of a surgical site infection across all types of surgery and that no cost-effectiveness data are presented. A cross-sectional study in 79 German hospitals studying 221 113 surgical patients has shown median case costs for patients with an SSI to be as high as €19 008 (approximately $25 200) compared with €9040 (approximately $12 000) for patients without an SSI.[3] Economic claims must be cautious—without formal cost-effectiveness analysis, the asserted thresholds are speculative.

As we can now safely assume that 85 RCTs, results from 2 trial sequential analyses, and a stable summary relative risk confirm the robustness of the data, future research needs to shift toward individual participant data meta-analyses in subgroups instead of new trials, standardized adverse-event reporting, and robust health-economic evaluations in clinical practice. The updated meta-analysis from Patel et al[1] forces the medical community to ask smarter questions about targeted use and value, rather than repeating broad efficacy trials, the subject of which the cumulative evidence has already largely settled. Only then can we answer the practical questions surgeons and health systems face: who benefits, under what protocol, and at what cost?

## ARTICLE INFORMATION

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# 1. iNPWT for prevention of SSI (JAMA Surgery 2026): the article you present

## Background you need

- **SSI types (CDC):** superficial incisional (skin, subcutaneous), deep incisional (fascia, muscle), organ/space (deeper anatomy, e.g. intra-abdominal abscess).
- **Wound class:** I clean, II clean-contaminated, III contaminated, IV dirty. In this review 93% were class I or II.
- **Devices:** PICO (Smith + Nephew, canister-free, about 80 mmHg) and PREVENA (3M, about 125 mmHg), usually 5 to 7 days. That is why pressure subgroups cluster at 80 and 125.
- **Mechanism (local):** removes exudate, reduces edema and lateral tension, improves edge apposition, protects the wound.
- **Prior evidence:** Groenen 2023 (57 RCTs, RR 0.67, TSA already conclusive). WHO, ACS/SIS and ASCRS give conditional recommendations for high-risk incisions. Two large null trials: WHIST (lower-limb fractures, JAMA 2020) and SUNRRISE (emergency laparotomy, JAMA 2025).
- **Methods vocabulary:** random effects with REML; $I^2$; leave-one-out; trial sequential analysis (is cumulative information enough to stop doing trials?); NNT/NNH; GRADE; AMSTAR 2 (quality of prior reviews).

## Key results

| Outcome | RR (95% CI) | Comment |
|---|---|---|
| Overall SSI (82 trials) | 0.64 (0.57 to 0.72) | 9.6% vs 14.4%; $I^2$ 31.6%; NNT 21 |
| Superficial SSI | 0.59 (0.49 to 0.72) | NNT 25 |
| Deep SSI | 0.66 (0.51 to 0.86) | NNT 74 |
| Organ/space SSI | 0.90 (0.67 to 1.20) | No effect: supports a local mechanism |
| Wound dehiscence / seroma / skin necrosis | 0.73 / 0.77 / 0.38 | All favor iNPWT |
| Readmission, mortality, SAE | Not significant | |
| Skin blistering | 4.51 (2.37 to 8.58) | 6.9% vs 1.3%; NNH 18 |
| Device-related adverse events | 11.87 (4.74 to 29.74) | 69/745 vs 0/823; NNH 11 |

## Strengths to defend

- RCTs only, PROSPERO registration, duplicate screening (kappa 0.94), RoB 2, GRADE, TSA, AMSTAR 2 appraisal of six prior reviews.
- Largest synthesis to date; first to report device-related adverse events separately.
- Superficial/deep benefit without organ/space benefit is biologically coherent and explains the null SUNRRISE result (infection there is driven by intra-abdominal contamination).
- Robust in leave-one-out and after excluding high risk-of-bias trials.

## Questions you will probably get, and how to answer

**Q: Does it make sense to pool cesarean, orthopedics and laparotomy?**

With 82 trials, an $I^2$ of 31.6% understates absolute dispersion, and no prediction interval is reported (it would be the honest answer about a new setting). Acknowledge it; point to the significant interaction by wound class and the null organ/space result as evidence that context matters.

**Q: WHIST and SUNRRISE were null. Is the effect driven by small studies?**

The funnel plot is asymmetric and Egger is not significant, but Egger has low power. Only 9 of 85 trials are low risk of bias; the sensitivity analysis removed only the 5 high-risk trials and kept 71 with 'some concerns'. Expect: why was GRADE not downgraded for publication bias?

**Q: Is NNT 21 generalizable?**

No. It comes from crude totals summed across trials (800/8338 vs 1207/8407), which ignores trial weighting, and it depends on a 14.4% baseline risk. In low-risk patients the NNT is much larger.

**Q: Is comparing NNT 21 with NNH 18 fair?**

An SSI and a blister do not weigh the same; this head-to-head comparison is the most attackable sentence. For device events the control arm had zero events, so the RR depends on a continuity correction; the risk difference is the right metric, and the outcome cannot occur by definition without a device.

**Q: Is the TSA convincing?**

The required information size was based on the observed effect rather than an a priori clinically relevant effect, which is close to circular (RIS 653 for overall SSI with 16,745 analyzed).

**Q: What about the invited commentary (Boermeester)?**

She says the message is not new after Groenen 2023 and argues against 'selective use' because an SSI costs about USD 25,000 (Germany). Answer: the paper's cost-neutral threshold (USD 3,150 to 10,500) varies with baseline risk, so 'selective' means risk-stratified, not restricted. Both sides agree the next step is IPD meta-analysis and formal cost-effectiveness work.

## Inconsistencies to know before someone finds them

- **[Error]** Table surgical specialty percentages do not match counts: 2842/8463 = 33.6%, Table says 32.3% (text, 33.5%, is right). Same for abdominal and trauma rows. (J tags)
- **[Error]** Device-related AE upper CI is 29.74 in the text and 29.73 in Figure 2.
- **[Method]** 'Adults' defined as 16 years or older.
- **[Method]** Class III/IV wounds are only about 7% of the sample: limited support for contaminated wounds.

***How to say it:*** *This is the largest and most rigorous synthesis to date. The effect on SSI is settled; the open questions are who benefits enough to justify blistering and cost, which needs IPD meta-analysis and cost-effectiveness data rather than more efficacy trials.*

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ORIGINAL ARTICLE
Breast

# Effectiveness of Intraoperative Ketorolac in Outpatient Breast Surgery: A Double-blinded Prospective Randomized Controlled Trial

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**Background:** The use of nonnarcotic analgesics, such as ketorolac, has been shown to reduce postoperative pain and opioid consumption. This double-blinded randomized trial is designed to assess the efficacy of intraoperative ketorolac in reducing postoperative narcotic use in outpatient breast reconstruction and reduction procedures.

**Methods:** This study is a prospective double-blinded randomized controlled trial. Adult patients, 18–64 years of age, undergoing breast surgery were randomized to receive 15 mg of ketorolac, 30 mg of ketorolac, or a placebo dose of saline. Patients' opioid requirements in the postoperative anesthesia care unit and postoperative opioid utilization and pain scores were collected through a daily survey. Postoperative hematomas were assessed before discharge and at subsequent follow-up visits for a period of 14 days.

**Results:** Of the 63 patients included in the study, 31 patients underwent delayed reconstruction following mastectomy and 35 patients underwent breast reduction surgery. Patients who received 30 mg of ketorolac had the fastest pain resolution ($P < 0.05$). The rate of opioid discontinuance was the fastest overall in patients who received 15 mg of ketorolac (rate = −0.072) when compared with the 30-mg ketorolac group (rate = −0.071) and the placebo group (rate = −0.065). Total opioid usage in the postoperative anesthesia care unit was not statistically different across the 3 groups. Only 1 patient developed a hematoma in the 15-mg ketorolac group.

**Conclusions:** This study demonstrates that a single dose of intraoperative ketorolac was associated with reduced opioid usage and postoperative pain. However, due to the study size, the difference in hematoma rate was not statistically significant.
*(Plast Reconstr Surg Glob Open 2025;13:e6540; doi: 10.1097/GOX.0000000000006540; Published online 17 February 2025.)*

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## INTRODUCTION

Despite national efforts to raise awareness and increase interventions, the opioid epidemic continues to cause significant injury and death in the United States. In 2019, there were approximately 10.1 million people who misused prescription opioids, 1.6 million people diagnosed with an opioid use disorder, and 48,006 people who died from a synthetic opioid-related overdose.[1] Despite these issues, prescription opioids remain an important component of postoperative pain management. Although efforts to support physicians in improving opioid prescribing patterns have helped some, prescription opioids, especially those prescribed as postoperative analgesics, continue to be a major contributor to the ongoing opioid crisis.[2] Currently, there are limited data regarding the rates of persistent opioid use after plastic and reconstructive operations. One retrospective cohort study looking at 466,677 patients undergoing 5 classes of plastic and reconstructive procedures (nasal, eye, breast, abdomen, and soft tissue reconstruction) found persistent opioid use, defined as opioid prescription fills between 90 and 180 postoperative days, occurred in 6.6% (30,865) of patients. Furthermore, patients undergoing breast (odds ratio 4.36;

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~~*ClinicalTrials.gov Identifier:*~~ *NCT0443939.*

K3

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95% confidence interval [CI] 4.10–4.63) and nasal (odds ratio 3.51; 95% CI 3.30–3.73) procedures had the greatest odds for persistent opioid use.[3]

The use of nonopioid analgesics as preventative analgesia has been recognized as a pathway to improve postoperative pain, while minimizing opioid use, in various surgical specialties.[4] Ketorolac, an intravenous nonsteroidal anti-inflammatory drug, has been a promising option for intraoperative analgesic control. In a meta-analysis study of 13 randomized controlled trials of 782 participants, a single dose of ketorolac was found to effectively prevent early postoperative pain and reduce opioid consumption compared with placebo.[4] Additionally, multiple studies on enhanced recovery after surgery (ERAS) protocols for breast surgery demonstrate the integral role of nonnarcotic medications when designing these protocols.[5–7] One recent study looking at ERAS protocols with Exparel in tissue expander-based breast reconstruction surgery found a decrease in postoperative opioid consumption. However, compared with Exparel, ketorolac provides a potentially equally beneficial but more cost-effective option to be incorporated into ERAS protocols.[6] Another retrospective review of 518 patients assessing multimodal pain management and opioid prescribing practices found that 15.6% and 25.7% of patients undergoing mastectomy with and without reconstruction received ketorolac intraoperatively, demonstrating a bias for ketorolac use in nonreconstruction cases and a need for further research to determine the safety and utility of ketorolac in breast surgery within plastic surgery.[7]

Despite these potential benefits, ketorolac use remains controversial in plastic surgery due to conflicting data suggesting an increased risk of postoperative bleeding and hematoma formation.[8] Concerns regarding ketorolac, a reversible cyclooxygenase inhibitor, stem from its mechanism of action surrounding the blockage of thromboxane A2 production and the potential to increase bleeding time and possible hematoma formation. One retrospective review of 379 patients undergoing breast reduction operation found that those who received ketorolac were at an increased risk of both surgical and non-surgical hematomas.[9] However, a separate study of 763 patients undergoing breast reduction and reconstruction procedures did not find such an association.[5] Another larger study of 981 patients also found no such association.[10] Of note, all previous studies assessing the use of ketorolac in breast surgery have been retrospective in nature.

There is still a paucity of data on the safety and efficacy of ketorolac in plastic surgery literature. To our knowledge, this is the first prospective double-blinded randomized controlled trial assessing the safety and efficacy of ketorolac in reducing postoperative pain and opioid usage after outpatient breast surgery.

## METHODS

This study is a prospective double-blinded randomized controlled trial. Approval was obtained from the Virginia Tech Carilion institutional review board and was registered on ClinicalTrials.gov (identification no. NCT04439396).

### Takeaways

**Question:** Can a single dose of ketorolac significantly decrease postoperative pain and opioid usage in outpatient breast surgery patients?

**Findings:** This double-blinded randomized controlled trial comparing 2 dosages of ketorolac and a placebo dose of saline in breast reconstruction and reduction surgery patients found that patients who received a single dose of 30 mg of ketorolac had significantly reduced postoperative pain and opioid usage.

**Meaning:** Plastic surgeons can reliably administer a single dose of 30 mg of ketorolac to reduce postoperative pain and opioid usage in breast surgery patients.

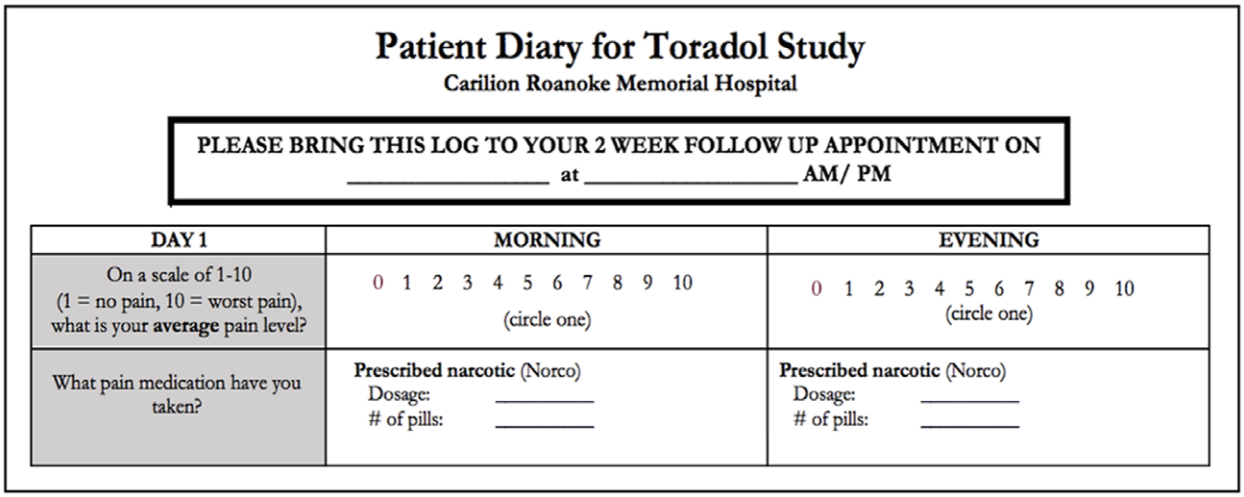
K4

Patients 18–64 years of age receiving outpatient breast surgery, including breast reduction, second-stage breast reconstruction, and minor breast revision surgery, were enrolled. All second-stage breast reconstruction procedures were expander-to-implant switch-out or implant revision procedures. Some patients did undergo simultaneous unilateral switch-out and contralateral reduction procedures. No patients undergoing mastectomy were included, as these patients had an overnight stay as part of their postoperative protocol. Patients with known renal disease, history of bleeding disorders, use of anticoagulation or antiplatelet drugs, allergy to ketorolac, or any known contraindication for ketorolac use were excluded.

The risks, benefits, and alternatives to participation in the study were explained to all patients before the procedure. Appropriate routine preoperative testing was completed for all patients to confirm surgical eligibility. A computer-generated randomization list was used to assign each patient to 1 of 3 groups; patients were assigned to receive 15 mg of ketorolac, 30 mg of ketorolac, or a placebo dose of saline. All study drugs were packaged identically and delivered to the anesthesiologist in the operating room on the day of the surgery. All patients received standard anesthesia medications according to each anesthesiologist's preferred protocol for intraoperative pain control in addition to the study drug. Additionally, no ketorolac was given intraoperatively for all patients. The study drug was administered by the anesthesiologist at the end of the case at the time of surgical site closure to maximize standardization between cases of variable time lengths. All operations were performed by 1 of 3 plastic surgeons at a single institution. For simplicity of study design, no patient received a local anesthetic block or postoperative nonnarcotic medications such as gabapentin during the study period.

All patients were monitored in the postoperative anesthesia care unit (PACU) and received standard postoperative pain management, including the administration of opioids, nonnarcotic pain medications, and antiemetics. However, no patients were given any additional ketorolac in the PACU period, regardless of study group designation. However, all medications administered in the PACU were not standardized and were instead at the discretion of each anesthesiologist. At discharge, patients were sent

K5



**Patient Diary for Toradol Study**

**Carilion Roanoke Memorial Hospital**

**PLEASE BRING THIS LOG TO YOUR 2 WEEK FOLLOW UP APPOINTMENT ON _____ at _____ AM/ PM**

| DAY 1 | MORNING | EVENING |
|---|---|---|
| On a scale of 1-10 (1 = no pain, 10 = worst pain), what is your **average** pain level? | 0 1 2 3 4 5 6 7 8 9 10 (circle one) | 0 1 2 3 4 5 6 7 8 9 10 (circle one) |
| What pain medication have you taken? | **Prescribed narcotic** (Norco) Dosage: _____ # of pills: _____ | **Prescribed narcotic** (Norco) Dosage: _____ # of pills: _____ |

**Fig. 1.** A graphic of the patient survey completed by patients at home regarding their postoperative pain and opioid usage.

home with a standardized number of 20 tablets of hydrocodone–acetaminophen 5–325 mg and a paper-form survey to track daily pain scores and opioid usage at home.

### Intraoperative Pain Medications

All patients in the study received standard-of-care intraoperative medications. Additionally, because anesthesiologists were also blinded to the study, no additional doses of ketorolac or other nonsteroidal anti-inflammatory drugs were administered other than the study drug, which was either a placebo dose of saline or either a 15 or 30 mg dose of ketorolac. However, other medications given intraoperatively were not standardized and were at the discretion of each anesthesiologist, within the guidelines of the standard of care. To determine if there were significant differences in intraoperative use of pain medications and narcotics across all 3 groups, an analysis of variance (ANOVA) test was performed to compare the use of these medications.

### Pain Assessment

Self-reported pain scores were collected twice daily using an 11-point numeric rating scale (Fig. 1). Although pain is a subjective, multidimensional phenomenon and difficult to objectively capture, multiple validated pain scales exist for clinical use. The numeric rating scale is a commonly used, validated pain scale for acute postoperative pain.[11]

### Opioid Consumption

Opioid usage in the immediate postoperative period while recovering in the PACU was collected through chart review. All opioids that were administered in the PACU were converted to morphine milligram equivalents (MMEs) for analysis. Patients were discharged with a standardized number of 20 tablets of hydrocodone–acetaminophen 5–325 mg. However, patients were advised to only take this medication as needed for pain.

Patients recorded their own opioid utilization by recording the number of prescribed pills consumed at home to manage their pain. Each patient's total bidaily dosage of opioids was calculated and converted to MMEs.

### Hematoma Formation

Evaluation for hematoma formation was conducted in the immediate postoperative period before discharge and again on the first postoperative follow-up at 1 week. All patients were discharged home with standardized postoperative care instructions, including guidelines on when to return to the clinic earlier than their first scheduled follow-up appointment or when to seek emergency medical attention.

### Statistical Analysis

A type III ANOVA test was performed to compare the rate of pain resolution over time and the rate of opioid usage over time to assess whether pain scores improved and opioid usage decreased differently across the 3 groups. To assess differences in total opioid usage in PACU, a one-way ANOVA test was conducted. All statistical analyses were conducted on RStudio and Microsoft Excel.

## RESULTS

A total of 63 patients were included in this study (Table 1). There were no statistical differences among study groups based on age, ethnicity, or procedure type. Patients who did not complete their patient diaries were excluded. Twenty-two participants were in the 15-mg ketorolac group, 30 were in the 30-mg ketorolac group, and 21 were in the placebo group. In the 15-mg ketorolac group, 12 (54.5%) patients underwent breast reductions and 10 (45.4%) underwent other reconstructive breast procedures. In the 30-mg ketorolac group, 12 (60.0%) patients underwent breast reductions and 8 (40.0%) patients underwent other breast reconstructive procedures. In the placebo group, 10 (47.6%) patients underwent breast reductions and 11 (52.4%) patients underwent other breast reconstructive procedures. All of the other breast reconstructive procedures were delayed, including

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K8

**Table 1. Patient Demographics Data**

| | 15 mg | 30 mg | Placebo | Total | Effect Size | *P* (Chi-squared) |
|---|---|---|---|---|---|---|
| Age, y | | | | | 0.1904 (small) | 0.3344 |
| 18–34 | 4 | 3 | 6 | 13 | | |
| 35–54 | 14 | 11 | 7 | 32 | | |
| 55–64 | 4 | 6 | 8 | 18 | | |
| Ethnicity | | | | | 0.2907 (moderate) | 0.0698 |
| White | 13 | 18 | 16 | 47 | | |
| African American | 9 | 2 | 5 | 16 | | |
| Procedure type | | | | | 0.0649 (small) | 0.8722 |
| Breast reduction | 12 | 12 | 10 | 34 | | |
| Breast reconstruction | 10 | 10 | 11 | 31 | | |

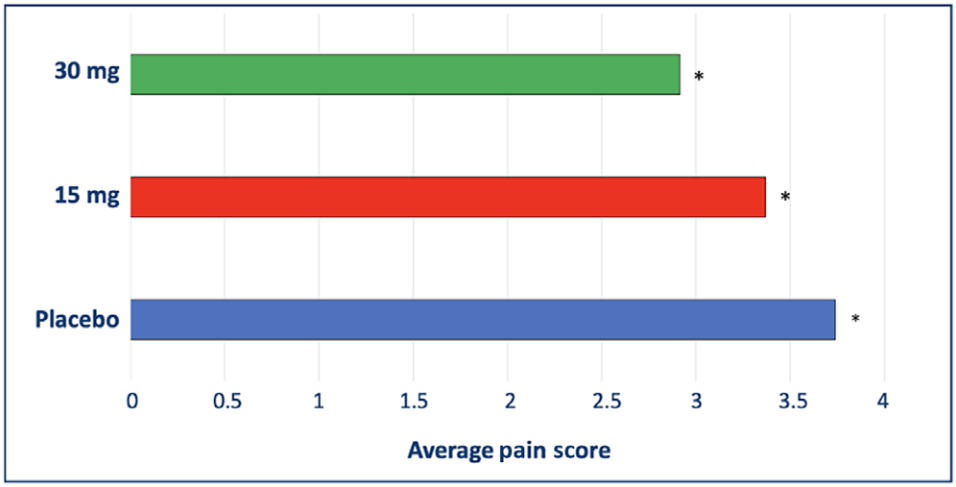



**Fig. 2.** A graph that shows average pain scores.

second-stage procedures such as expander-to-implant switch-out operations or implant revisions. Patient demographics are shown in Table 1.

### Pain Scores

Average pain scores were calculated from the patient surveys. Patients in the 30-mg ketorolac group had the lowest pain scores out of the 3 groups with an average pain score of 2.92. Patients in the placebo group had the highest pain scores out of the 3 groups with an average pain score of 3.74. Patients in the 15-mg ketorolac group had an average pain score of 3.37 (Fig. 2). There was a statistically significant difference across the 3 groups when using an ANOVA test at the *P* value less than 0.05 level ($P = 0.0001$).

When looking at the rate of pain improvement over the postoperative period, there was a statistically significant difference across all 3 groups ($P < 0.001$). Patients in the 30-mg ketorolac group had the fastest rate of pain resolution, and patients in the placebo group had the slowest (Fig. 3). Post hoc analysis determined that the 30-mg ketorolac group's rate of pain improvement was statistically faster than the 15-mg ketorolac and placebo group (Table 2).

### Opioid Usage After Discharge

Postdischarge opioid utilization was calculated by converting total narcotic consumption to MMEs. Patients in the 30-mg ketorolac group had the lowest opioid usage with an average MME of 1.13. Patients in the placebo group had the highest opioid usage with an average MME of 1.41. Patients in the 15-mg group had an average MME of 1.20 (Fig. 4). There was a statistically significant difference across the 3 groups when using an ANOVA test at the *P* value less than 0.05 level ($P = 0.0004$).

An ANOVA test was conducted to evaluate whether opioid usage after discharge changed over time. Although results did not demonstrate a statistically significant difference between the 3 groups ($\chi^2 = 0.378$, $P = 0.828$), the 15-mg ketorolac group had the fastest rate of opioid discontinuance (−0.0717) and the placebo group had the slowest (−0.0651) (Fig. 5). When comparing the rate of change in opioid discontinuance, there was no a statistically significant difference between the groups (Table 2).

### Intraoperative Pain Medications

Other than the study drug, only 6 pain medications were administered across all 3 groups intraoperatively. These included acetaminophen, lidocaine, ketamine, fentanyl, hydromorphone, and morphine. The average dosages of all 6 medications are shown in Table 3. An ANOVA test was performed to compare the use of acetaminophen, lidocaine, ketamine, and total MME. This showed no statistical difference in other administered medications across the 3 groups (Table 3).

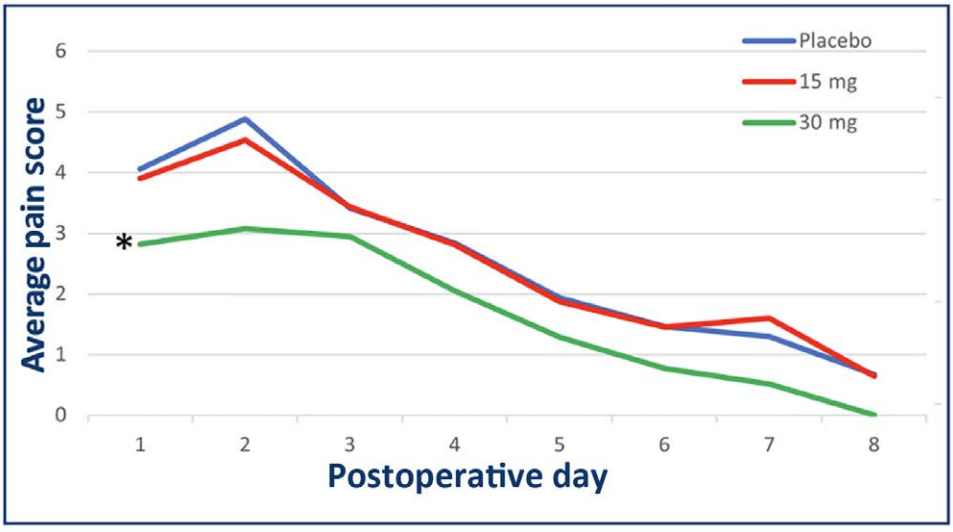



**Fig. 3.** A graph that shows the comparison of rate of pain resolution.

**Table 2. Analysis of Rates of Pain Resolution**

| | | ANOVA | | | |
|---|---|---|---|---|---|
| | Rate of Pain Improvement | Comparison Group | Estimate | SE | *P* |
| 15 mg | −0.267 | Versus 30 mg | 0.118 | 0.0289 | <0.0001* |
| 30 mg | −0.385 | Versus placebo | −0.2127 | 0.0306 | <0.0001* |
| Placebo | −0.172 | Versus 15 mg | −0.0947 | 0.0314 | 0.0026 |

*$P < 0.05$. SE, standard error.

K12

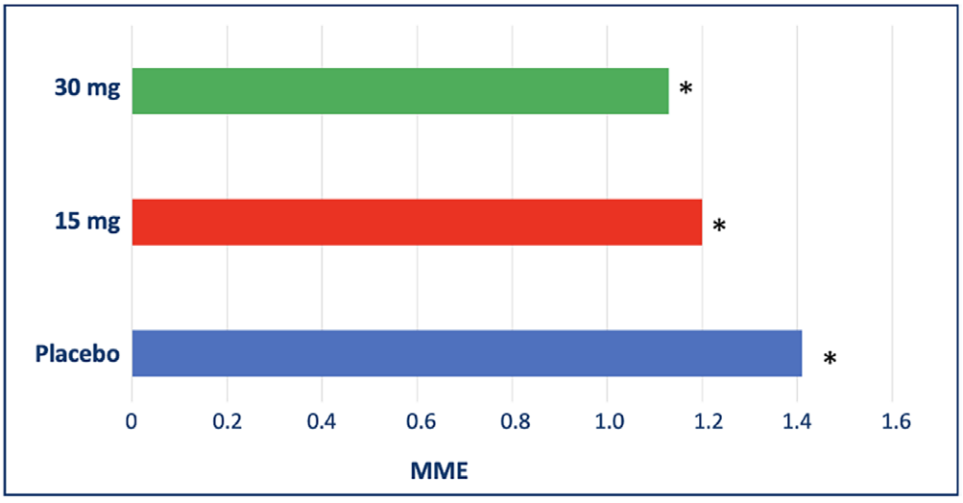



**Fig. 4.** A graph that shows the average opioid consumption (MME).

**PACU Narcotic Requirements**

To compare the total opioid usage in PACU, total narcotic usage was first converted to MME. A one-way ANOVA comparing total MME usage in PACU did not find a statistical difference among the 3 groups [$F(2,63) = 2.40$, $P = 0.09$]. Although these data currently lack statistical significance with our limited sample size, post hoc analysis suggests that there may be some differences across the groups (Table 4). Although not statistically significant, patients in the 15- and 30-mg ketorolac groups had the largest difference of 77.3 and the lowest *P* value of 0.085.

K13

**Complications**

Of the total enrolled patients in the study, only 1 patient developed a hematoma (15-mg ketorolac group), for an overall rate of 1.5%. This patient underwent a breast reduction procedure which was uneventful in the initial postoperative period. However, on postoperative day 10,

K14

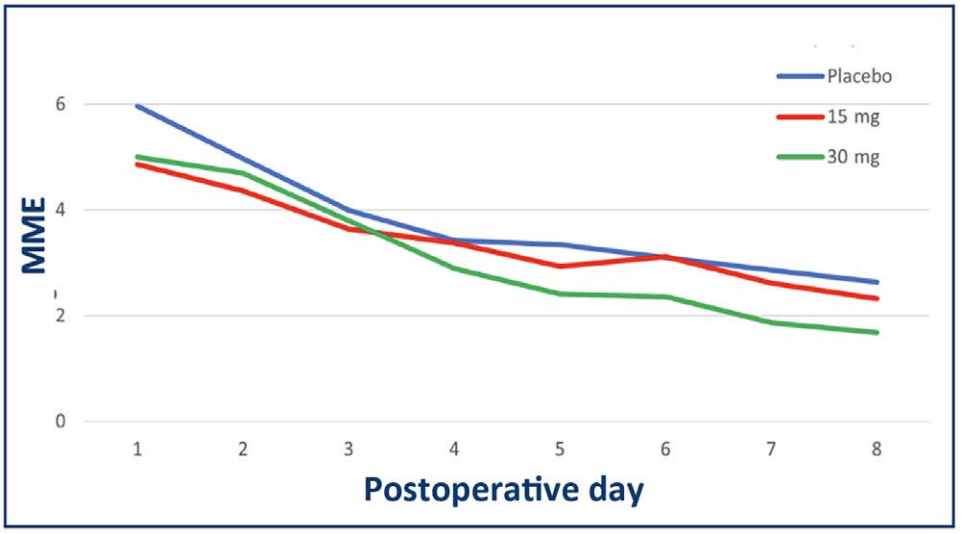



**Fig. 5.** A graph that shows the comparison of rate of opioid discontinuance.

**Table 3. Average Doses and Analysis of Intraoperative Pain Medications Using ANOVA**

| | Acetaminophen (mg) | Lidocaine (mg) | Ketamine (mg) | Fentanyl (mg) | Hydromorphone (mg) | Morphine (mg) | Total MME |
|---|---|---|---|---|---|---|---|
| 15 mg | 750 | 76.6 | 2.00 | 0.125 | 0.575 | 0.2 | 49.6 |
| 30 mg | 632 | 85.2 | 7.37 | 0.132 | 0.237 | 0.0 | 44.2 |
| Placebo | 684 | 79.0 | 4.74 | 0.111 | 0.368 | 0.5 | 42.1 |
| *f* | 0.308 | 0.297 | 1.143 | — | — | — | 1.09 |
| *P* | 0.736 | 0.744 | 0.326 | — | — | — | 0.345 |

**Table 4. Analysis of Opioid Discontinuance**

| | | ANOVA | | | |
|---|---|---|---|---|---|
| | Rate of Opioid Discontinuance | Comparison Group | Estimate | SE | *P* |
| 15 mg | −0.0717 | Versus 30 mg | −0.0008 | 0.0106 | 0.9390 |
| 30 mg | −0.0709 | Versus placebo | −0.0057 | 0.0112 | 0.6089 |
| Placebo | −0.0651 | Versus 15 mg | −0.0066 | 0.0115 | 0.5688 |

SE, standard error.

the patient developed a delayed hematoma, which was evacuated surgically. The patient then recovered appropriately with no other complications.

Given our small sample size and incidence rate of hematomas, we were unable to assess for significant hematoma rates associated with ketorolac. Using a 2-sided, large-sample *z* test of the Poisson event-rate difference at a significance level of 0.050 (95% CI), the analysis of a total sample size of 63 subjects, each measured for a fixed duration, achieves ≥40.99% power and would have been able to detect a difference of 50% between the ketorolac group event rate of 1.50 and the placebo group event rate of 1.00. No patient developed acute renal failure or other bleeding complications such as a gastrointestinal bleed or cerebral hemorrhage during the study period.

## DISCUSSION

Ketorolac is a powerful short-acting analgesic frequently used by multiple surgical specialties since its approval for medical use in 1989. Its relatively low cost and ease of intraoperative administration make it an affordable and accessible option for pain control. Previous studies have demonstrated the effective analgesic and opioid-sparing effects of ketorolac use in various surgical settings, including in orthopedics, general surgery, and gynecology.[4] Other studies have specifically investigated ketorolac use in plastic surgery.[9] However, all previous studies looking at ketorolac use in breast surgery have been retrospective in nature and, therefore, limited in their conclusions. Furthermore, existing literature provides conflicting results on the safety of ketorolac use in breast surgery. Some studies have suggested an increased risk of hematoma formation from ketorolac use following major breast operations, whereas other studies have failed to reproduce this effect.[8,10] These contradicting conclusions are further muddied by the retrospective nature of these studies.

Before addressing the uncertainty surrounding the hematoma risk associated with ketorolac use, it is important that its effectiveness at pain reduction be fully

established. This study sought to determine the effect of a single dose of intraoperative ketorolac on postoperative pain and subsequent opioid reduction.

Our results showed that patients who received 30 mg of ketorolac reported the lowest average pain scores and lowest average opioid usage overall across the 3 groups. These patients also demonstrated the fastest rate of pain resolution and the second fastest rate of opioid discontinuance, demonstrating that a single intraoperative dose of ketorolac is effective at decreasing postoperative pain and reducing opioid usage. This is of particular importance considering the lack of existing data on opioid stewardship and prescribing patterns within plastic surgery.[3] One study of 466,677 patients found that among a cohort of opioid-naive adults undergoing plastic and reconstructive operations, a significant portion of them demonstrated prolonged and persistent opioid use postoperatively.[3] Best practice guidelines surrounding opioid prescription patterns for patients undergoing plastic and reconstructive operations are still lacking. Therefore, it is imperative that plastic surgeons understand their options for pain control to decrease postoperative opioid usage in their patients.

Only 1 patient enrolled in the study developed a hematoma. Although the primary purpose of this study was not to evaluate the hematoma rate, it is of obvious interest. A sample size large enough to detect small changes in the hematoma rate given the relative rarity of the complication would be quite large and was beyond the scope of this study. We were able to demonstrate that there was not an extremely large increase (50% or greater), which is consistent with the available data on ketorolac use. Future studies of large sample populations may be able to tease out if there is a small increase in risk associated with ketorolac.

In addition to the relatively small sample size, another limitation of this study included the subjectivity of pain and variability of pain tolerance among the general population. For example, patients undergoing breast reconstruction surgery after mastectomy have experienced a prior breast operation and may have a higher tolerance or increased awareness of what to expect in subsequent breast operations, whereas patients undergoing breast reduction surgery may have never had prior breast surgery. However, variances such as these would be controlled for by randomization as seen by the even distribution of breast reduction versus breast reconstruction cases in each group and within the study. Additionally, intraoperative and postoperative pain medications administered were at the discretion of each anesthesiologist, resulting in a nonhomogenous group of patients' pain control. However, as this variability was equally applicable to all groups, post hoc analysis did not find a statistically significant difference in pain medications administered.

## CONCLUSIONS

Opioid stewardship within the field of plastic surgery is an ongoing issue. This study shows that a single intraoperative dose of ketorolac in outpatient breast surgery significantly reduces postoperative pain and opioid consumption. These results provide support for the efficacy of ketorolac in the outpatient surgery setting and the potential for consideration of ketorolac in ERAS protocols. Future studies with larger sample sizes would be needed to detect small changes in hematoma rates in these patients.

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## DISCLOSURE

*The authors have no financial interest to declare in relation to the content of this article.*

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# 2. Intraoperative ketorolac in outpatient breast surgery (PRS Global Open 2025)

## Background

- **Drug:** IV NSAID, reversible COX inhibitor; lowers thromboxane A2 and platelet aggregation, which is the basis of the hematoma concern. Doses 15 or 30 mg; half-life about 5 hours.
- **The controversy:** retrospective studies conflict (Cawthorn 2012 found more hematomas after reduction; Nguyen 2018, Rojas 2019 and Stephens 2015 did not). Gobble 2014 (27 RCTs, all specialties) found no excess bleeding.
- **Recent synthesis:** Almeida et al., Aesthetic Plastic Surgery 2025: 7 studies (6 observational plus this RCT), ketorolac in breast reduction, hematoma OR 2.63 (1.58 to 4.37), although crude incidence was slightly lower with ketorolac (6.1% vs 6.7%).
- **Units:** pain on an 11-point NRS (MCID about 1 to 2 points). MME: hydrocodone factor 1, so one Norco 5/325 tablet = 5 MME.

## What it reports

Patients randomized to ketorolac 15 mg, 30 mg or saline at closure. Mean pain 2.92 (30 mg), 3.37 (15 mg), 3.74 (placebo); post-discharge MME 1.13, 1.20, 1.41. One hematoma (15-mg group).

## Problems, with the evidence

| | Issue | Evidence in the paper |
|---|---|---|
| E | Four different totals | Abstract 63 patients but 31 + 35 = 66; Results 22 + 30 + 21 = 73; Table 1 34 + 31 = 65 (K1, K4) |
| E | 30-mg group size | Reported as 30, but 12 (60%) + 8 (40%) = 20; 22 + 20 + 21 = 63, so '30' is likely a typo |
| E | Text vs table | 8 reconstructions in the 30-mg group in text, 10 in Table 1 |
| E | Population contradiction | Abstract: 'reconstruction following mastectomy'; Methods: no mastectomy patients included |
| E | Registry number | NCT0443939 on page 1 vs NCT04439396 in Methods |
| E | Analgesia contradiction | Methods: no postoperative non-narcotic drugs; PACU paragraph: non-narcotic pain medications given |
| E | Statistics | F(2,63) should be F(2,60); an 'ANOVA' reported with a chi-square statistic; 15 mg vs placebo P = 0.0026 not marked significant; hematoma 1/63 = 1.6%, not 1.5% |
| M | Not intention to treat | Patients without diaries excluded after randomization; numbers per arm and CONSORT diagram absent |
| M | No primary outcome or sample size | Multiple outcomes, no a priori power calculation |
| M | Repeated measures | 'Type III ANOVA' used for daily scores; a mixed model was needed |
| M | Clinical relevance | 0.8-point pain difference is below the MCID; 0.28 MME is a fraction of a pill |
| M | Biological plausibility | A single dose with a 5-hour half-life is unlikely to change pain on day 7 |
| M | Hematoma overclaim | 41% power and 1 event cannot exclude a 50% increase, yet the authors say they demonstrated it |
| M | Takeaway | 'Plastic surgeons can reliably administer 30 mg' goes well beyond the data |

***How to say it:*** *The question matters and an RCT is the right design, but internal inconsistencies, post-randomization exclusions and effect sizes below clinical relevance mean this trial cannot settle either efficacy or hematoma risk.*

## Is a new meta-analysis viable?

A narrow review (ketorolac and hematoma in breast reduction) has just been published (Almeida 2025). Room remains for (1) a broader review across all breast procedures pooling efficacy and safety by procedure; (2) a methodologically better analysis (adjusted estimates only, RCT and observational separated, risk differences and NNH, rare-event methods, dose and timing); (3) a Bayesian analysis of the probability of a clinically important absolute increase in hematoma. Check first: status of Cochrane protocol CD013290, PROSPERO registrations, and a scoping count of eligible studies (target about 15 or more).

# Hormone replacement before brain death increases organ donation rate after catastrophic brain injury: An EAST multicenter trial

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After reading the featured articles published in the *Journal of Trauma and Acute Care Surgery*, participants should be able to demonstrate increased understanding of the material specific to the article. Objectives for each article are featured at the beginning of each article and online. Test questions are at the end of the article, with a critique and specific location in the article referencing the question topic.

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**BACKGROUND:** One third of organ donors die due to catastrophic brain injury (CBI). CBI causes massive physiological derangements that can damage organs and prevent organ donation. Hormone replacement therapy (HRT) can mitigate this damage when given in the critical care setting, but no standardized protocol for use at trauma centers exists. Single-center studies show that HRT increases organ donation, but there has not been an investigation on which hormone combinations are most beneficial.

**METHODS:** This prospective, observational EAST multicenter trial included adult CBI patients at 38 Level I and II trauma centers from January 2022 to June 2025. CBI was defined as a brain injury causing loss of function above the brain stem and subsequent death. A generalized linear mixed model analyzed the association of HRT and organ donation.

**RESULTS:** Of 1,184 CBI patients, 704 received at least one hormone. There were 269 organ donors, and 227 received HRT. HRT patients were younger, more often male, sustained higher rates of penetrating brain injuries, and had lower rates of known comorbidities compared with non-HRT patients. HRT patients had higher ISS, but there was no difference in abbreviated injury scale-head scores between groups. GLMM analysis accounted for demographic and clinical differences between groups and found that single-hormone replacement with steroids, levothyroxine, vasopressin, dopamine, DDAVP, or insulin was associated with increased odds ratios of organ donation compared with no HRT. The combination of methylprednisolone and levothyroxine was associated with the greatest increased odds of donation.

**CONCLUSIONS:** When administered to CBI patients before brain death, HRT is associated with significantly increased odds of organ donation. A standardized protocol incorporating HRT should be utilized across trauma centers to maximize organ donation in patients who have suffered nonsurvivable head trauma. (*J Trauma Acute Care Surg*. 2026;101: 619–631. Copyright © 2026 Wolters Kluwer Health, LLC. All rights reserved.)

**LEVEL OF EVIDENCE:** Prognostic and Epidemiological; Level II.

**KEY WORDS:** Catastrophic brain injury, traumatic brain injury, brain death, organ donation, organ transplant.

There is significant variation in the treatment of catastrophic brain injury (CBI), defined as traumatic brain injury expected to result in permanent loss of brain function above the level of the brain stem.[1] Some trauma centers have institutional protocols for the management of CBI patients, but these vary from one hospital to another.[2–5] In addition, many hospitals do not have protocols, and care varies between providers. Regardless, the goals of treatment may include stabilizing patients to be reunited with family before death and preserving the option of organ donation.[1] One third of organ donors die due to CBI, and these trauma patients are generally young with few comorbidities, and thus have high organ donation potential.[6] CBI can cause brain stem herniation and ischemia, which often leads to vasoregulatory, endocrine, and metabolic derangements, multiorgan failure and cardiovascular collapse.[1,7]

These insults to transplantable organs can have negative impacts on donation and transplant. Hormones used to minimize physiological disturbance and organ damage in CBI patients who are potential organ donors include vasopressin, DDAVP, steroids, levothyroxine, insulin, and dopamine.[2,3,5,8–15] The evidence that many institutional CBI protocols cite supporting hormone replacement therapy (HRT) comes from data on brain-dead organ donors or animal models of brain death. This research appropriately supports donor management protocols (many of which include HRT) for brain-dead organ donors used by organ procurement organizations (OPOs).[8,16–19] While these studies and OPO protocols show the benefit of HRT after brain death, there are few studies investigating the utility of HRT for CBI before brain death occurs.

Single-center studies have found that institutional CBI protocols, including HRT, are associated with increased organ donation rates.[3,5,9,16] Prior results from this EAST multicenter trial investigating the relationship of institutional CBI protocols and organ donation found that the presence of a CBI protocol itself does not correlate with organ donation, suggesting that some protocols hold more utility than others.[4] We did find that administration of either steroids or vasopressin was associated with increased odds of organ donation.[4] Here we focus more in-depth on the utility of HRT. The objective of this study is to investigate which hormones, when re-

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placed either alone or in combination, in CBI patients before brain death, are associated with organ donation, and which combinations of hormones are most beneficial for preserving organ donation potential.

## METHODS

This prospective, observational, multicenter trial sponsored by the Eastern Association for the Surgery of Trauma includes 38 Level I and II trauma centers in the United States, Greece, and Saudi Arabia. Institutional review board approval was obtained by each site, and waivers of consent were granted. STROBE guidelines ensured proper reporting of methods, results, and discussion (Supplemental Digital Content [SDC] 1, http://links.lww.com/TA/F569). Data were collected via *REDCap*. Patients were included if they presented to a participating center between January 1, 2022, and June 15, 2025. The Joint Trauma System definition for CBI was used: "brain injury that is expected after imaging evaluation and/or clinical exam to result in the permanent loss of all brain function above the brain stem level, based on assessment by the trauma surgeon, neurosurgeon, or other intensivist."[1] Inclusion criteria were adult (18+ years old) trauma patients who suffered a traumatic mechanism CBI and expired in-hospital because of brain death or circulatory death caused by their head injury. Exclusion criteria were patients younger than 18 years and those with nontraumatic CBI (eg, spontaneous hemorrhage, ischemic stroke). Pregnant and incarcerated patients were not excluded if they met the inclusion criteria, but these characteristics were not collected. Organ donation data were collected from hospital records and from the associated OPO when possible.

### Study Cohort, Primary and Secondary Outcomes

H2

Patients were categorized into an HRT group (received at least one of the following: methylprednisolone, vasopressin, DDAVP, levothyroxine, dopamine, insulin) or a non-HRT group. The primary outcome was the organ donation rate. The secondary outcome was tissue donation rate. Exploratory outcomes included organ donation rates by specific hormones and hormone combinations administered.

### Study Variables

Demographic and clinical data collected included age, sex, race, known medical comorbidities, organ donor registry status, injury type, injury severity score (ISS), abbreviated injury scale prehospital timing intervals, prehospital and emergency department (ED) vital signs, and medical management, including hormone administration. Admission vital signs collected included Glasgow Coma Scale (GCS), systolic blood pressure (SBP), diastolic blood pressure, heart rate (HR), and respiratory rate. Hospital course variables included length of stay (LOS), intensive care unit (ICU) LOS, ventilator days, location of death, type of death (circulatory or brain death), and in-hospital complications, including acute kidney injury (AKI), acute respiratory distress syndrome, and venous thromboembolism. Organ and tissue donation variables included donor registry status, whether next-of-kin was approached for organ/tissue donation and whether they consented, whether donation occurred, and which organs/tissues were donated. Participating centers contacted their associated OPO for detailed information on donor management and donation information.

### Statistical Analysis

#### Descriptive Statistics

Categorical variables were reported as frequencies and percentages; continuous variables as median with interquartile range given non-normal distributions confirmed by Shapiro-Wilk testing (all $p < 0.05$). Univariate comparisons between HRT and non-HRT groups used Mann-Whitney $U$ tests for continuous variables and $\chi^2$ or Fisher exact tests for categorical variables, based on expected cell frequencies.

#### Generalized Linear Mixed Model

A generalized linear mixed model (GLMM) was used to analyze the association between the administration of hormones and organ donation. For the primary GLMM, complete-case analysis after listwise deletion of 272 patients missing ≥ 1 covariate yielded a final sample of 799 patients across 29 centers. Covariates for the model were selected using three converging methods: (1) a directed acyclic graph constructed from clinical domain knowledge with application of the backdoor criterion,[20] (2) the disjunctive cause criterion (variables associated with either exposure or outcome at $p < 0.20$),[21] and (3) change-in-estimate analysis (≥ 10% change in the exposure odds ratio). Variables identified by at least two of three methods were retained. The final adjustment set comprised 13 covariates: age, sex, ISS, ED GCS, Marshall CT classification, registered donor status, do-not-resuscitate order, self-inflicted injury, presence of institutional CBI protocol, comorbidities, and race (Caucasian, Black/African American, Hispanic/Latino). Full covariate selection process detailed in SDC 2, Supplemental Digital Content 2, http://links.lww.com/TA/F570. Multicollinearity was assessed using variance inflation factors < 5 for all covariates. Missing data were not imputed; sensitivity analyses using multiple imputation confirmed robustness of results. Detailed patient flow and missing data patterns reported in SDC 3, Supplemental Digital Content 3, http://links.lww.com/TA/F571 and 4, Supplemental Digital Content 4, http://links.lww.com/TA/F572.

Administration of individual hormones was analyzed using a GLMM with center as a random intercept among the 799 complete cases. Co-administration of two or three hormones was analyzed using multivariable lo- H3 gistic regression among patients receiving at least the specified hormones; combination groups are nonexclusive. All models adjusted for 13 covariates: age, sex, ISS, ED GCS, Marshall CT classification, registered organ donor status, do-not-resuscitate order, self-inflicted injury, presence of institutional CBI protocol, co-


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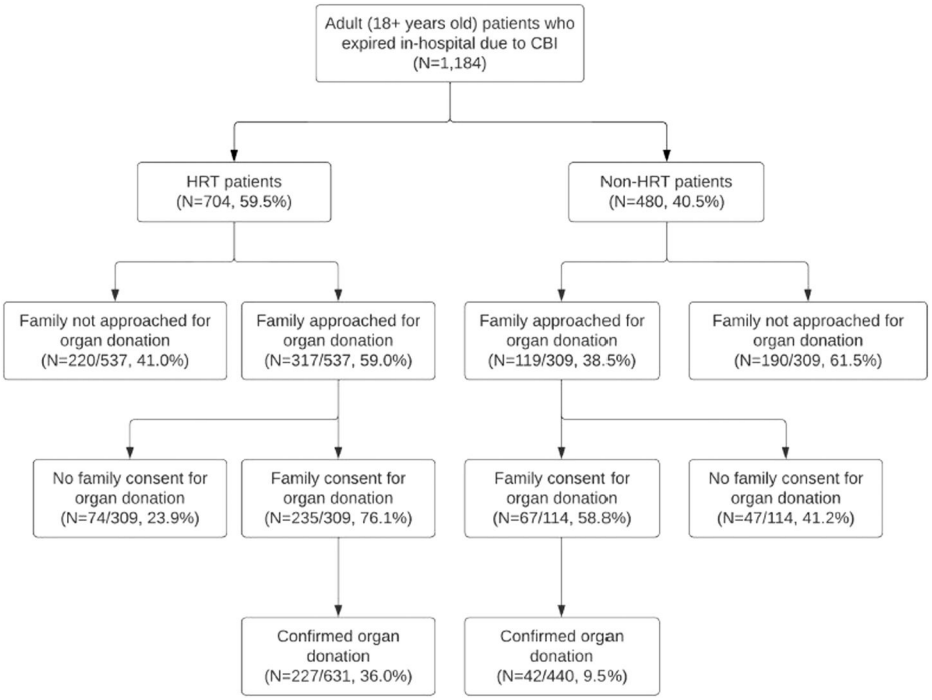



**Figure 1.** CONSORT flow diagram of CBI patients who presented to trauma centers and expired in-hospital and organ donation outcomes.

morbidities, and race. Point estimates were plotted on a logarithmic scale; point size is proportional to the precision (inverse variance) of the estimate. Combinations were included if the sample size was ≥75, false discovery rate–corrected $p < 0.05$, and CI ratio <10.

The GLMM had a binomial distribution and logit link function, fitted using the R package lme4 (version 1.1-38).[22] Center was modeled as a random intercept to account for within-center correlation arising from heterogeneous CBI protocols, OPO workflows, and institutional donation cultures across the sites. The intraclass correlation coefficient was calculated to quantify the proportion of variance in organ donation attributable to center-level differences. Model discrimination was assessed using the C-statistic (area under the receiver operating characteristic curve), and calibration was evaluated using the Brier score and the Hosmer-Lemeshow goodness-of-fit test. Results were reported as adjusted odds ratios (aOR) with 95% CI.

## Post Hoc Power Analysis

For the GLMM results, because the sample size was fixed by the completed trial enrollment (N = 799), a post hoc power analysis was conducted to characterize the detectable effect sizes rather than to inform study design prospectively. Using the Hsieh-Bloch-Larsen formula for logistic regression with binary exposure,[23] the minimum detectable effect size (MDES) at 80% power (two-sided $\alpha = 0.05$) was OR = 1.97, given N = 799 (483 HRT, 316 no-HRT; baseline donation rate 8.2% in the unexposed group). For a conservatively small effect (OR = 1.5), the study would not be adequately powered (37%); however, the observed adjusted effect was substantially larger (aOR = 4.60, 95% CI: 2.65–7.96), exceeding the MDES by 2.3-fold. Power calculations used the crude odds ratio, which is standard practice for sample size estimation.

For hormone combinations with smaller sample sizes, the inclusion threshold of N ≥75 was formally justified by power analysis: at N = 75 (assuming balanced exposure), the MDES is OR = 6.88, ensuring that only combinations with adequate power to detect moderate-to-large effects were reported. Combination-specific power analysis confirmed that all 8 combinations meeting this threshold were individually adequately powered (power ≥92%). Sample size requirements for future randomized controlled trials were calculated for both the observed crude effect (N = 94) and a conservative effect size of OR = 3.0 (N = 252). This analysis is reported with the caveat that post hoc power calculations have recognized limitations.[24] Analyses performed using Python 3.13.5 with SciPy 1.16.1and scikit-learn packages, and R 4.5.1 with lme4 (version 1.1-38) and mediation packages.[25–27]

# RESULTS

## Demographic and Injury Characteristics

A total of 1,184 CBI patients were included in this analysis (Fig. 1). At least one hormone was administered to

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**TABLE 1.** Demographic and Injury Characteristics

| | All Patients (N = 1184) | HRT Patients (N = 704) | Non-HRT Patients (N = 480) | *p* |
|---|---|---|---|---|
| Demographics | | | | |
| Age (y) | 53.0 (31.0–74.0) | 42.0 (28.0–64.0) | 67.0 (41.0–81.0) | < 0.001 |
| Missing | 10 (0.8) | 3 (0.4) | 7 (1.5) | — |
| Sex | | | | |
| Male | 847 (71.5) | 537 (76.3) | 310 (64.6) | < 0.001 |
| Race/ethnicity | | | | |
| Arab | 15 (1.3) | 13 (1.8) | 2 (0.4) | 0.058 |
| Asian | 26 (2.2) | 16 (2.3) | 10 (2.1) | 0.987 |
| African | 3 (0.3) | 2 (0.3) | 1 (0.2) | 1.000 |
| Black/African American | 176 (14.9) | 122 (17.3) | 54 (11.2) | 0.005 |
| Caucasian | 668 (56.4) | 343 (48.7) | 325 (67.7) | < 0.001 |
| Hispanic/Latino | 164 (13.9) | 122 (17.3) | 42 (8.8) | < 0.001 |
| American Indian | 9 (0.8) | 9 (1.3) | 0 (0.0) | 0.013 |
| Pacific Islander | 5 (0.4) | 4 (0.6) | 1 (0.2) | 0.654 |
| Other | 42 (3.5) | 29 (4.1) | 13 (2.7) | 0.259 |
| Unknown | 88 (7.4) | 59 (8.4) | 29 (6.0) | 0.163 |
| Missing | 7 (0.6) | 1 (0.1) | 6 (1.2) | — |
| Known comorbidities | | | | |
| None | 332 (28.0) | 234 (33.2) | 98 (20.4) | < 0.001 |
| Atrial fibrillation | 88 (7.4) | 34 (4.8) | 54 (11.2) | < 0.001 |
| Cancer | 73 (6.2) | 29 (4.1) | 44 (9.2) | < 0.001 |
| Chronic obstructive pulmonary disease | 66 (5.6) | 28 (4.0) | 38 (7.9) | 0.006 |
| End-stage renal disease | 28 (2.4) | 18 (2.6) | 10 (2.1) | 0.740 |
| Congestive heart failure | 68 (5.7) | 30 (4.3) | 38 (7.9) | 0.012 |
| Diabetes | 163 (13.8) | 91 (12.9) | 72 (15.0) | 0.352 |
| Liver disease | 31 (2.6) | 15 (2.1) | 16 (3.3) | 0.277 |
| Injury characteristics | | | | |
| Mechanism of injury | | | | |
| Blunt | 904 (76.4) | 519 (73.7) | 385 (80.2) | 0.012 |
| Penetrating | 229 (19.3) | 158 (22.4) | 71 (14.8) | 0.001 |
| Mixed | 20 (1.7) | 15 (2.1) | 5 (1.0) | 0.231 |
| Other | 15 (1.3) | 11 (1.6) | 4 (0.8) | 0.403 |
| Missing | 16 (1.4) | 1 (0.1) | 15 (3.1) | |
| Intent of injury | | | | |
| Assault | 108 (9.1) | 74 (10.5) | 34 (7.1) | 0.056 |
| Accident | 801 (67.7) | 458 (65.1) | 343 (71.5) | 0.025 |
| Self-inflicted | 144 (12.2) | 91 (12.9) | 53 (11.0) | 0.377 |
| Unknown | 95 (8.0) | 69 (9.8) | 26 (5.4) | 0.009 |
| Injury severity | | | | |
| ISS, median (IQR) | 27.0 (25.0–35.0) | 29.0 (25.0–38.0) | 26.0 (25.0–34.0) | 0.030 |
| AIS-head | 5.0 (4.0–5.0) | 5.0 (4.0–5.0) | 5.0 (4.0–5.0) | 0.051 |
| AIS-face | 2.0 (1.0–2.0) | 2.0 (1.0–2.0) | 2.0 (1.0–2.0) | < 0.001 |
| AIS-thorax | 3.0 (2.0–3.0) | 3.0 (2.0–3.0) | 3.0 (2.0–3.0) | 0.161 |
| AIS-abdomen | 2.0 (2.0–3.0) | 2.0 (2.0–3.0) | 2.0 (2.0–3.0) | 0.614 |
| AIS-extremity | 2.0 (1.0–3.0) | 2.0 (2.0–3.0) | 2.0 (1.0–3.0) | 0.207 |

Continuous variables presented as median (IQR). Binary variables presented as n (%).
AIS, Abbreviated Injury Scale.

704 (59.5%) of these patients, who constituted the HRT cohort, and 480 (40.5%) patients received no HRT (Table 1). Patients who received HRT were younger than those who did not [median (interquartile range) age 42 (28–64) years old vs. 67 (41-81) years old, $P < 0.001$], and more often males (76.3% vs. 64.6%, $p < 0.001$). More HRT patients had no known comorbidities compared with non-HRT patients (33.2% vs. 20.4%, $p < 0.001$). Patients who received HRT had lower rates of atrial fibrillation (4.8% vs. 11.2%, $p < 0.001$), cancer (4.1% vs. 9.2%, $p < 0.001$), and chronic obstructive pulmonary disease (4.0% vs. 7.9%, $p = 0.006$). Overall, most patients had a blunt mechanism of injury (76.4%), and about a quarter (22.4%) of patients sustained a penetrating injury. However, HRT patients had


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**TABLE 2.** Pre-Hospital and Emergency Department Characteristics

| | All Patients (N = 1184) | HRT Patients (N = 704) | Non-HRT Patients (N = 480) | *p* |
|---|---|---|---|---|
| Prehospital care | | | | |
| Time from 911 call to EMS arrival (min) | 11.0 (7.0–21.0) | 11.0 (7.0–20.0) | 12.0 (7.0–25.5) | 0.095 |
| EMS Transport time (min) | 27.0 (18.0–39.0) | 27.0 (18.0–40.0) | 28.0 (18.0–39.0) | 0.722 |
| Prehospital vital signs | | | | |
| GCS | 3 (3–8) | 3 (3–7) | 3 (3–10) | 0.005 |
| SBP (mmHg) | 138 (106–162) | 134 (105–160) | 142 (109–175) | 0.044 |
| DBP (mmHg) | 81 (62–98) | 81 (62–97) | 81 (64–101) | 0.839 |
| MAP (mmHg) | 102 (81–122) | 101 (79–121) | 106 (87–123) | 0.140 |
| HR (beats/min) | 88 (68–108) | 90 (69–110) | 84 (66–103) | 0.004 |
| RR (breaths/min) | 16 (12–22) | 18 (12–22) | 16 (12–20) | 0.120 |
| Temp (°F) | 98.0 (96.7–98.5) | 98.0 (96.8–98.6) | 97.8 (96.5–98.3) | 0.551 |
| Emergency department vital signs | | | | |
| GCS | 3 (3–7) | 3 (3–5) | 3 (3–9) | < 0.001 |
| SBP (mmHg) | 130 (100–155) | 128 (100–149) | 134 (105–160) | 0.018 |
| DBP (mmHg) | 78 (60–95) | 76 (60–95) | 80 (62–94) | 0.272 |
| MAP (mmHg) | 97 (78–116) | 97 (76–116) | 99 (80–116) | 0.383 |
| HR (beats/min) | 94 (74–114) | 98 (77–118) | 89 (71–108) | < 0.001 |
| RR (breaths/min) | 18 (16–22) | 19 (16–22) | 18 (15–22) | 0.003 |
| Temp (°F) | 97.3 (96.0–98.0) | 97.3 (95.9–98.1) | 97.3 (96.1–98.0) | 0.754 |

Continuous variables presented as median (IQR).
EMS, emergency medical services; DBP, diastolic blood pressure; MAP, mean arterial pressure; RR, respiratory rate; Temp, temperature.

a higher rate of penetrating trauma (22.4% vs. 14.8%, $p = 0.001$) and a lower rate of blunt trauma (73.7% vs. 80.2%, $p = 0.012$) and accidental injury (65.1% vs. 71.5%, $p = 0.025$) compared with non-HRT patients. HRT patients also had higher ISS [29 (25–38) vs. 26 (25–34), $p = 0.030$] but similar head Abbreviated Injury Scale (5 vs. 5, $p = 0.051$) compared with non-HRT patients.

## Prehospital and ED Arrival Characteristics

There was no difference in transport times (27 vs. 28 min, $p = 0.722$) between cohorts (Table 2). While the range of prehospital GCS was statistically lower in the HRT cohort, median GCS was 3 in both cohorts [3 (3–7) vs. 3 (3–10), $p = 0.005$]. HRT patients had lower SBP on-scene (134 mm Hg (105–160) vs. 142 mm Hg (109–174), $p = 0.044$), but there was no difference in median arterial pressure between groups (101 mm Hg (79–121) vs. 106 mm Hg (87–123), $p = 0.140$). The HRT cohort had higher prehospital HR [90 beats per minute (BPM) (69–110) vs. 84 BPM (66–103), $p = 0.004$].

On trauma bay arrival, HRT patients had statistically lower GCS [3 (3–5) vs. 3 (3–9), $p < 0.001$], lower SBP [128 mm Hg (100-149) vs. 134 mmHg (105–160), $p = 0.018$], higher HR [98 BPM (77–118) vs. 89 BPM (71–108), $p < 0.001$], and higher respiratory rate [19 breaths per minute (16–22) vs. 18 breaths per minute (15–22), $p = 0.003$] compared with non-HRT patients.

## Outcomes

All patients suffered fatal head injuries and died due to the head trauma. Of the 1,140 patients with a known type of death (circulatory or brain death), the majority (646/1140, 56.7%) died due to circulatory death (Table 3). HRT patients progressed to brain death at nearly three times the rate of non-HRT patients (58.7% vs. 22.4%, $p < 0.001$), and had a significantly lower rate of circulatory death (41.3% vs. 77.6%, $p < 0.001$).

In causal mediation analysis, brain death was identified as a mediator, rather than a confounder, on the causal pathway (HRT→brain death→donation) to organ donation. Approximately one third of the HRT effect on organ donation was mediated through brain death progression (Average Causal Mediation Effect = 0.061; proportion mediated = 33.3%; 95% CI: 14.6%–51.7%), while two-thirds represented a direct effect independent of brain death status (Average Direct Effect = 0.097; $p < 0.001$).

HRT patients had longer hospital LOS than non-HRT patients [3.0 (1.0–6.0) days vs. 2.0 (1.0–6.2) days, $p = 0.002$], and ICU LOS [3.0 (1.0–6.0) days vs. 1.5 (0.0–6.0) days, $p < 0.001$]. Most patients (74.3%) expired in the ICU, with HRT patients expiring at a higher rate in the ICU than non-HRT patients (84.0% vs. 59.2%, $p < 0.001$), and at a lower rate in the ED (4.1% vs. 16.5%, $p < 0.001$). Overall, complications that could impact organ donation were infrequent, but HRT patients had significantly higher rates of AKI (15.8% vs. 9.8%, $p = 0.004$) and acute respiratory distress syndrome (4.5% vs. 1.5%, $p = 0.006$), with no difference in venous thromboembolism (1.6% vs. 1.7%, $p = 1.000$).

## Organ Donation

Of 1,184 patients in the study, 1,071 (90.5%) had confirmed organ donation outcomes, and 269 (25.1% of patients with known donation outcomes, 22.5% of all

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**TABLE 3.** In-hospital Outcomes

| | All Patients (N = 1184) | HRT Patients (N = 704) | Non-HRT Patients (N = 480) | *p* |
|---|---|---|---|---|
| Circulatory death | 646/1140 (56.7) | 283/685 (41.3) | 319/411 (77.6) | < 0.001 |
| Brain death | 494/1140 (43.3) | 402/685 (58.7) | 92/411 (22.4) | < 0.001 |
| Unknown | 88 | 19 | 69 | |
| Length of stay | | | | |
| Hospital LOS, d (n = 1147) | 3.0 (1.0–6.0) | 3.0 (1.0–6.0) | 2.0 (1.0–6.2) | 0.002 |
| ICU LOS, d (n = 1143) | 2.0 (1.0–6.0) | 3.0 (1.0–6.0) | 1.5 (0.0–6.0) | < 0.001 |
| Death location | | | | |
| Emergency department | 103 (9.0) | 29 (4.1) | 74 (16.5) | < 0.001 |
| Operating room | 61 (5.3) | 42 (6.0) | 19 (4.2) | 0.248 |
| ICU | 854 (74.3) | 589 (84.0) | 265 (59.2) | < 0.001 |
| Floor | 79 (6.9) | 26 (3.7) | 53 (11.8) | < 0.001 |
| Other | 46 (4.0) | 12 (1.7) | 34 (7.6) | < 0.001 |
| Unknown | 41 (3.5) | 6 (0.8) | 35 (7.3) | |
| Complications | | | | |
| AKI | 158 (13.3) | 111 (15.8) | 47 (9.8) | 0.004 |
| ARDS | 39 (3.3) | 32 (4.5) | 7 (1.5) | 0.006 |
| VTE | 19 (1.6) | 11 (1.6) | 8 (1.7) | 1.000 |
| Family consent for donation | | | | |
| Family approached for organ donation | 436/846 (51.5) | 317/537 (59.0) | 119/309 (38.5) | < 0.001 |
| Family approached for tissue donation | 292/571 (51.1) | 213/353 (60.3) | 79/218 (36.2) | < 0.001 |
| Family consented to organ donation | 302/423 (71.4) | 235/309 (76.1) | 67/114 (58.8) | < 0.001 |
| Family consented to tissue donation | 199/279 (71.3) | 153/204 (75.0) | 46/75 (61.3) | 0.037 |
| Donation outcomes | | | | |
| Organ donation | 269/1071 (25.1) | 227/631 (36.0) | 42/440 (9.5) | < 0.001 |
| Tissue donation | 114/1017 (11.2) | 90/600 (15.0) | 24/417 (5.8) | < 0.001 |

Continuous variables presented as median (IQR). Binary variables presented as n (%).
ARDS, acute respiratory distress syndrome; VTE, venous thromboembolism.

patients) donated organs. Surrogate decision makers of HRT patients were approached by an OPO representative for organ donation more often (59.0% vs. 38.5%, $p < 0.001$), and they more frequently gave consent for organ donation (76.1% vs. 58.8%, $p < 0.001$) compared with non-HRT patient surrogates. HRT patients successfully donated organ(s) more often than non-HRT patients (36.0% vs. 9.5%, $p < 0.001$).

A higher percentage of organ donors received methylprednisolone (51.3% vs. 5.3%), vasopressin (74.8% vs. 31.3%), DDAVP (39.5% vs. 12.6%), levothyroxine (43.7% vs. 7.7%) (all $p < 0.001$), and dopamine (8.9% vs. 3.9%, $P = 0.003$) compared with nondonors (Table 4). Insulin was administered at similar rates to donors and nondonors.

## Tissue Donation

Tissue donation was analyzed for the 1,017 patients with confirmed tissue donation outcomes. Fewer patients donated tissues compared with organs; of the 1,071 patients, 114 (10.6% of patients with known donation outcomes, 9.6% of all patients) donated tissue(s). Similarly to organ donation, surrogate decision makers of HRT patients were also approached for tissue donation more often (60.3% vs. 36.2%, $p < 0.001$), and they gave consent for tissue donation at a higher rate (75.0% vs. 61.3%, $p = 0.037$) than non-HRT surrogate decision makers. HRT patients donated tissue(s) more frequently (15.0% vs. 5.8%, $p < 0.001$) than non-HRT patients.

Also, similarly to organ donors, tissue donors received methylprednisolone (34.2% vs. 14.0%, $p < 0.001$), vasopressin (69.0% vs. 38.2, $p < 0.001$), DDAVP (29.7% vs. 17.2%, $p = 0.002$), and levothyroxine (34.8% vs. 14.1%, $P < 0.001$) at higher rates compared with nondonors. There was no difference in the dopamine or insulin administration rate between groups.

## Hormone Replacement Variability

Of the 38 participating centers, 17 hospitals had CBI protocols, all of which included at least one hormone. Of the 38 hospitals in this study, 35 (92.1%) administered hormone(s) to at least one patient; 16 of the 17 centers (94.1%) with protocols and 19 of the 21 centers (90.5%) without protocols gave some form of HRT. Of institutional protocols, 14 (82.4%) included vasopressin, 10 (58.8%) levothyroxine, 9 (52.9%) steroids, 9 (52.9%) DDAVP, 7 (41.2%) dopamine, and 6 (35.3%) insulin. Triggers for administering the same hormone differed between protocols; for example, of the ten centers with levothyroxine in their protocol, five initiate therapy for all CBI patients, while the other five only give it to patients requiring vasopressors. Dosing for hormones also varied;


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**TABLE 4.** Hormone Replacement and Organ/Tissue Donation

| Organ Donation | All Patients All Patients (N = 1071) | Donors Organ Donors (N = 269) | Nondonors Organ Nondonors (N = 802) | *p* |
|---|---|---|---|---|
| Methylprednisolone | 177/1055 (16.8) | 135/263 (51.3) | 42/792 (5.3) | < 0.001 |
| Vasopressin | 448/1062 (42.2) | 199/266 (74.8) | 249/796 (31.3) | < 0.001 |
| DDAVP | 203/1054 (19.3) | 103/261 (39.5) | 100/793 (12.6) | < 0.001 |
| Levothyroxine | 175/1054 (16.6) | 114/261 (43.7) | 61/793 (7.7) | < 0.001 |
| Dopamine | 54/1050 (5.1) | 23/259 (8.9) | 31/791 (3.9) | 0.003 |
| Insulin | 267/1046 (25.5) | 72/258 (27.9) | 195/788 (24.7) | 0.353 |
| Tissue Donation | All Patients (N = 1017) | Tissue Donors (N = 114) | Tissue Nondonors (N = 903) | |
| Methylprednisolone | 163/1003 (16.3) | 38/111 (34.2) | 125/892 (14.0) | < 0.001 |
| Vasopressin | 420/1009 (41.6) | 78/113 (69.0) | 342/896 (38.2) | < 0.001 |
| DDAVP | 186/1003 (18.5) | 33/111 (29.7) | 153/892 (17.2) | 0.002 |
| Levothyroxine | 165/1004 (16.4) | 39/112 (34.8) | 126/892 (14.1) | < 0.001 |
| Dopamine | 52/1000 (5.2) | 7/110 (6.4) | 45/890 (5.1) | 0.723 |
| Insulin | 256/994 (25.8) | 35/111 (31.5) | 221/883 (25.0) | 0.173 |

Binary variables presented as n (%). DDAVP- Desmopressin

for example, some centers give loading doses of levothyroxine (either 10 or 20 µg) followed by a drip, while other centers only give a drip (ranging from 10–40 µg/h). At centers without protocols, hormones are administered, and their dosages also vary depending on providers' practices.

## GLMM for Hormones Associated With Successful Organ Donation

In the GLMM, administration of individual hormones was associated with increased odds of organ donation (Fig. 2). Methylprednisolone had the strongest association [aOR: 19.28 (95% CI: 10.23–36.32), $p < 0.001$], followed by levothyroxine [aOR: 7.72 (4.27–13.95), $p < 0.001$], hydrocortisone [aOR: 4.41 (1.23–15.78), $p < 0.001$], vasopressin [aOR: 3.88 (2.41–6.26), $p < 0.001$], dopamine [aOR: 3.48 (1.53–7.95), $p = 0.003$], DDAVP [aOR 3.39 (2.09–5.50), $p < 0.001$], and insulin [aOR: 3.33 (1.18–9.37), $p = 0.023$].

Several combinations of two hormones were associated with increased odds of organ donation: methylprednisolone and levothyroxine [aOR: 22.89 (11.04–47.47)], methylprednisolone and vasopressin [aOR: 17.05 (9.25–31.43)], methylprednisolone and DDAVP [aOR: 15.55 (7.23–33.47)], vasopressin and levothyroxine [aOR: 9.95 (5.61–17.63)], DDAVP and levothyroxine [aOR: 9.714 (0.55–20.71)], and vasopressin and DDAVP [aOR: 5.75 (3.36–9.82)] ($p < 0.001$ for all).

Co-administration of three hormones was associated with increased odds of donation compared with no HRT: methylprednisolone, vasopressin, and levothyroxine [aOR: 21.85 (10.35–46.09)] and methylprednisolone, vasopressin, and DDAVP [aOR: 14.66 (6.46–33.27)] ($p < 0.001$ for both). No significant pairwise interactions were detected among the hormone pairs (all $p > 0.46$), suggesting additive rather than synergistic effects.

## DISCUSSION

This EAST-sponsored multicenter trial investigated the utility of HRT for CBI treatment before brain death. There is no civilian equivalent of the military's Joint Trauma System CBI management protocol (which includes HRT[1]), so there is great variation in the current treatment of CBI patients. While trauma centers may have their own protocols, these vary. At centers without protocols, practices vary between providers in terms of which, if any, hormones are administered. Here, we aimed to investigate which hormones, administered alone or together, are associated with increased odds of organ donation.

Patients who received HRT were younger, more often male, and suffered increased rates of penetrating head trauma compared with non-HRT patients. This may reflect demographic differences at centers that employ HRT for CBI patients compared with centers that do not. The fact that treatment differed for younger, majority non-white male patients and older white patients does indicate that current treatment of CBI patients may not be equitable, and underscores the need for standardized treatment. After controlling for population differences between HRT and non-HRT patients, our results demonstrate increased adjusted odds ratios of organ donation with the use of HRT before brain death. The combination of methylprednisolone and levothyroxine had the highest overall odds of organ donation compared with other HRT strategies.

While organ donation can occur following circulatory or brain death, or as a gift from a living donor, brain-dead organ donors are the largest group of organ donors (over 7,000 of the 20,000 donors in 2025) in the United States.[28] In this study, most CBI patients who received HRT died due to brain death, rather than circulatory death. In the setting of CBI and its ensuing physiological disruptions that can cause rapid deterioration, the higher rate of progression to brain death may suggest the role of HRT in restoring homeostasis and

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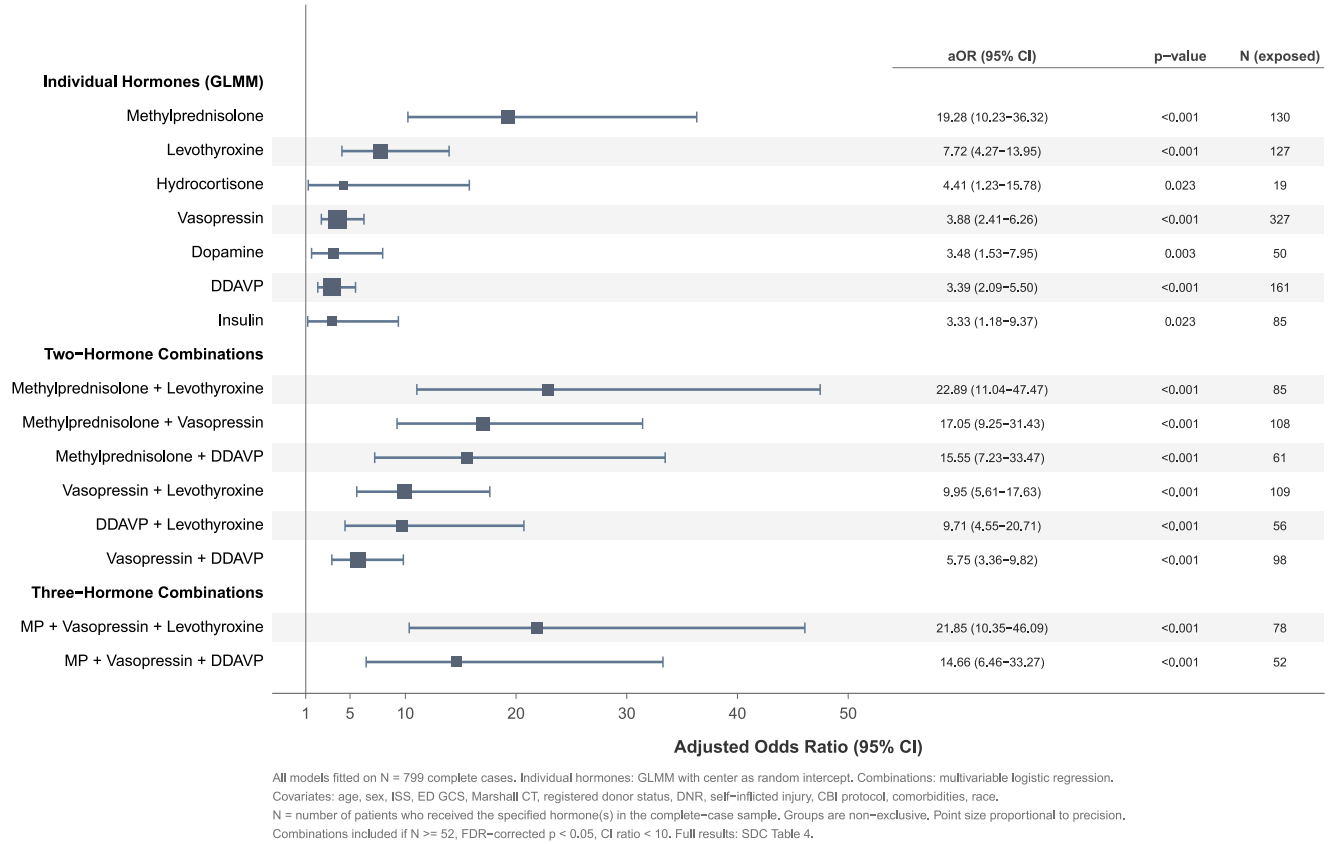



**Figure 2.** Forest plot of adjusted odds ratios (aOR) with 95% CI for the association between hormone replacement therapy and organ donation in patients with catastrophic brain injury. Individual hormones were analyzed using a generalized linear mixed model (GLMM) with center as a random intercept among 799 complete cases across 29 centers. Two and three hormone combinations were analyzed using multivariable logistic regression among patients receiving at least the specified hormones; combination groups are nonexclusive.

preventing cardiovascular collapse and subsequent circulatory death. In animal models and brain-dead human patients, hormone administration stabilizes hemodynamic instability and mitigates organ damage, and increases organ donation.[9,18,29–32] Our causal mediation analysis found brain death is a mediator, and not a confounder, of organ donation. This supports the possibility that HRT preserves organ viability through the aforementioned mechanisms when administered to CBI patients before brain death.

In concordance with prior single-center studies investigating the impact of protocolized HRT for CBI patients in the critical care setting before brain death,[2,3,5,8,16,33] we found increased rates of organ donation with HRT. Prior studies have investigated various combinations of hormones replaced, including methylprednisolone, insulin, and levothyroxine with or without steroids.[2,3,5,8,33,34] Past studies have found that each of these HRT approaches is associated with increased organ donation rate, but have not been able to compare strategies to identify the most impactful treatment.

Corticosteroids aid in vascular tone regulation, enhance response to catecholamine vasopressors, treat adrenal insufficiency, and attenuate the inflammatory effects of cytokines released after massive brain injury and brain death.[35–39] Treatment with methylprednisolone after brain death can dampen this inflammatory response and allow successful transplant of lungs that would otherwise be considered unacceptable for transplant.[40] Co-administration of methylprednisolone and levothyroxine can significantly reduce vasopressor requirements in brain-dead organ donors, thereby mitigating the negative impacts of pressors on transplantable organs.[16,34] Levothyroxine alone has shown mixed results in terms of effect on successful organ donation and transplant.[9,10,41,42] Hyperglycemia is involved with inflammation, endothelial dysfunction, and increased blood-brain barrier permeability after brain injury.[9,11,33] Insulin administration for hyperglycemia in brain-dead organ donors helps attenuate inflammatory response and has been associated with improved renal function after transplant.[11,43,44] Similarly, increased renal perfusion with low-dose dopamine after brain death improves renal graft function and long-term outcomes.[12,13] There is relatively scant literature on the use of DDAVP in potential organ donors.[45,46] Vasopressin,


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while often used as a vasopressor, can also be used to treat diabetes insipidus, and also reduces catecholamine pressor requirements, systemic inflammatory response, and pulmonary injury, and has been shown to increase perfusion and graft function of kidneys, hearts, and lungs.[14,47,48] While it is a long-acting synthetic vasopressin analogue, DDAVP is a V2 agonist with higher affinity for volume status modulation and water balance with minimal cardiovascular effect.[49] DDAVP is also used to treat diabetes insipidus, which is common after brain injury and brain death. Its use can lead to AKI and, in some studies, has been shown to decrease renal donation potential.[45,50]

This multicenter study compared multiple HRT approaches with different hormone combinations, which represent standardized in institutional CBI protocols as well as practice patterns at centers without such protocols. Interestingly, GLMM analysis found that co-administration of three hormones was not associated with a higher odds of organ donation compared with co-administration of two hormones. The combination of methylprednisolone and levothyroxine had the highest odds of donation. At the seven centers that included methylprednisolone in their CBI protocol, four administered to all patients, two administered to only patients requiring vasopressors, and one center's protocol did not specify. Of the 10 centers that included levothyroxine in their CBI protocol, 5 administered it as a continuous infusion to all patients and 5 used drips for hypotensive patients. Six centers gave a 20 mcg bolus before the drip.

H16 ~~This study has several limitations.~~ As is true with any retrospective trial, there is the potential for coding errors and/or missing data. While we included data on organ and tissue donation outcomes using information available in hospital charts, centers had variable success in obtaining more detailed donation information from their associated OPO. Similar to how trauma centers have different CBI management practices, OPOs have their own donor management protocols that vary from one to another. Variation in donor management after transfer to the OPOs may have impacted donation outcomes; most centers were unable to obtain much data on this donor management. In addition, there was variability in hormone dosage and frequency, both written into protocols and in clinical practice. Given the substantial exposure variability in patient care across the 38 centers, it is likely that we have incomplete confounder control. Future prospective investigation with more subjects will be needed to determine the optimal dosing of hormones to maximize the benefit of HRT. There were also some demographic differences between the HRT and no-HRT cohorts. This may be explained by patient demographic differences at hospitals with and without CBI protocols and was controlled for in multivariate analysis, although it could signal additional confounders that were not accounted for. In addition, the study lacked granular data points, including continuous physiology, laboratory data, computed tomography imaging, echocardiogram, bronchoscopy, and other clinical information that may impact organ donation and transplant. Due to the aforementioned variable success in collaborating with the OPOs associated with each center, the study largely lacks detailed information on short-term and long-term transplant outcomes. Despite numerous limitations, this study is strengthened by its multicenter nature and large study population and is the largest study on the topic of CBI management.

## CONCLUSIONS

This prospective, observational multicenter trial, including 38 Level I and II centers in the United States, Greece, and Saudi Arabia, found that HRT with individual hormones and combinations ~~of dopamine~~, DDAVP, vasopressin, hydrocortisone, levothyroxine, and methylprednisolone was associated with increased odds of organ donation. Furthermore, treatment of CBI patients with methylprednisolone and levothyroxine was associated with the greatest odds of successful organ donation. Routine administration of methylprednisolone and levothyroxine to CBI patients in the ICU may hold significant clinical utility. Standardized CBI management, including hormone replacement, appears warranted and could positively impact organ donation.

H14

H15

### AUTHORSHIP

K.D.N., S.T., and A.P. participated in conception and study design. K.D.N. participated in the literature review and drafting of the manuscript. M.B.P., A.P., J.C.D., J.N., J.A.K., A.N.H., A.K., R.N., A.W.M., A.B.P., B.W., J.S., P.O.U., A.S., C.S., A.E., J.M., B.P., E.I., B.M., E.C., R.M.B., S.A.Z., B.H.S., K.N.H., J.Z., T.E., I.A., P.M., J.M., M.S.F., M.D.P., C.T., P.S., J.D.S., T.M., L.E.J., J.M.W., J.D.S., P.K., R.H., D.G., A.S, C.V.B., J.V., S.M.C., F.A., J.M.H., K.L., J.A.M., D.R., D.T., E.T., T.H.J., G.D., A.C.B., B.S., B.D., S.L.M., N.P., R.S., S.V., B.S.M., M.A.W., K.M., S.P., S.N.L., M.H., V.D., C.M., T.J.M., N.B., W.A., C.A., R.J.J., R.C.H., M.J.L., H.M., R.P.D., C.A.F., W.T.HT., Y.T., V.M., F.M., J.D.B., D.R.M., C.H.P., M.B.A., N.M.G., J.F., and A.A.S. participated in data acquisition. S.H. and L.G. participated in data analysis and interpretation. K.D.N., S.T., L.G., S.H., J.A.K., and A.H. critical revision.

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### DISCLOSURE

All JTACS Disclosure forms have been supplied and are provided as Supplemental Digital Content, http://links.lww.com/TA/F573.

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# 3. Hormone replacement before brain death (J Trauma Acute Care Surg 2026, EAST)

## Background

- **Catastrophic brain injury (CBI):** non-survivable injury expected to progress to loss of all brain function above the brainstem. Patients are often young and healthy, so good donor candidates.
- **Physiology:** herniation causes a catecholamine storm, then vasoplegia and hypothalamic-pituitary failure. Loss of ADH causes diabetes insipidus (vasopressin, DDAVP); low thyroid hormone impairs the heart (levothyroxine); adrenal insufficiency and inflammation (methylprednisolone, hydrocortisone); hyperglycemia (insulin). Dopamine is a pressor grouped with them.
- **Donation routes:** after brain death (DBD) or after circulatory death (DCD).
- **OPO (Organ Procurement Organization):** the federally designated nonprofit (about 55 in the US) that evaluates potential donors, approaches families for consent, manages the donor after brain death (including hormones by its own protocol) and allocates organs through UNOS/OPTN. Brazilian equivalent: OPO within the Sistema Nacional de Transplantes.
- **Prior evidence:** mostly from brain-dead donors (Novitzky, 63,593 donors). An RCT of levothyroxine in unstable brain-dead donors (NEJM 2023) did not increase hearts transplanted. The novelty here is starting HRT before brain death, modeled on the military Joint Trauma System CBI guideline.
- **Methods vocabulary:** GLMM with center as random intercept (accounts for hospital culture); ICC; covariates by DAG/backdoor criterion; causal mediation (ACME = indirect effect via brain death, ADE = direct effect).

## What it reports

1,184 CBI deaths at 38 centers. 704 received at least one hormone. Donation 36.0% with HRT vs 9.5% without; aOR for any HRT 4.60 (2.65 to 7.96). Individual hormones: methylprednisolone aOR 19.28, levothyroxine 7.72, hydrocortisone 4.41, vasopressin 3.88, dopamine 3.48, DDAVP 3.39, insulin 3.33. Highest combination: methylprednisolone + levothyroxine aOR 22.89. Authors recommend routine methylprednisolone + levothyroxine.

## Limitation 1: confounding by donation intent (the main problem)

HRT is not given at random. Teams start it when they have already decided a patient is a donor candidate, so HRT may be a marker of the donation pathway rather than its cause.

| | HRT | No HRT |
|---|---|---|
| Median age | 42 years | 67 years |
| No comorbidities | 33.2% | 20.4% |
| Family approached by OPO | 59.0% | 38.5% |
| Family consented | 76.1% | 58.8% |
| Donated organs | 36.0% | 9.5% |

The model adjusts for age, comorbidities, donor registration and DNR, but cannot adjust for what was not measured: clinician intent, timing of OPO referral and intensity of ICU management. **Direction: inflates the HRT effect.**

## Limitation 2: immortal time bias

To receive HRT, a patient must live long enough to be given it. Those who die early can only be in the no-HRT group, which makes that group look worse for reasons unrelated to hormones.

| Where or how death occurred | HRT | No HRT |
|---|---|---|
| Emergency department | 4.1% | 16.5% |
| Floor | 3.7% | 11.8% |
| ICU | 84.0% | 59.2% |
| Median ICU stay | 3.0 days | 1.5 days |
| Circulatory death | 41.3% | 77.6% |

Analogy: 'patients who received a second chemotherapy cycle lived longer'; they had to survive the first cycle to get the second. Correct approach: HRT as a time-varying exposure, or a landmark analysis (e.g. patients alive at 24 hours, exposure classified up to then). **Direction: inflates the HRT effect.**

## Limitation 3: timing of HRT is unknown (reverse causation)

The title says 'before brain death', but the HRT group is anyone who received a hormone at any time. After brain death is declared, the OPO takes over and usually gives methylprednisolone and levothyroxine to nearly every donor.

| Sequence the authors assume | Sequence that may have happened |
|---|---|
| 1. Severe brain injury | 1. Severe brain injury |
| 2. ICU gives hormones | 2. Brain death declared |
| 3. Patient stays stable | 3. Family consents |
| 4. Brain death declared | 4. OPO gives hormones (donor protocol) |
| 5. Donation | 5. Donation |
| Hormone causes donation | Donation pathway causes hormone use (reverse causation) |

Supporting data (Table 4): 135 of 177 methylprednisolone recipients (76%) became donors; 51.3% of donors vs 5.3% of non-donors received it. Levothyroxine: 43.7% vs 7.7%. Analogy: 'patients who received the discharge kit survived more'; only survivors get the kit. This can explain why methylprednisolone has the largest aOR. **Direction: strongly inflates the effect, especially for methylprednisolone and levothyroxine.**

## Limitation 4: odds ratios look bigger than the real effect

With a common outcome (25% donate), the OR overstates the relative risk. Approximate conversion (Zhang and Yu, baseline 8.2% in the no-HRT group):

| Exposure | Reported aOR | Approximate RR |
|---|---|---|
| Any HRT | 4.60 | about 3.5 |
| Methylprednisolone | 19.28 | about 7.7 |

Still large, but a single drug producing an effect this big in an observational study should itself raise suspicion of residual confounding.

## Limitation 5: the 'best combination' claim is not supported

| Strategy | aOR vs no HRT (95% CI) | N exposed |
|---|---|---|
| Methylprednisolone alone | 19.28 (10.23 to 36.32) | 130 |
| Methylprednisolone + levothyroxine | 22.89 (11.04 to 47.47) | 85 |
| Methylprednisolone + vasopressin + levothyroxine | 21.85 (10.35 to 46.09) | 78 |

### 5a. No head-to-head comparison

Every combination is compared only with no HRT, never with the others. Ranking requires A + B vs A and vs B. Analogy: three runners each beat a slow walker; that does not tell you which runner is fastest. The intervals above overlap almost completely, and adding a third drug gives a lower point estimate.

### 5b. Overlapping groups

Groups are non-exclusive: a patient on methylprednisolone + vasopressin + levothyroxine is counted in at least four bars. 78 of the 85 'methylprednisolone + levothyroxine' patients (92%) also received vasopressin, which is why the two ORs are nearly identical (22.89 vs 21.85). The effect of levothyroxine cannot be separated from that of vasopressin.

### 5c. Center effect dropped

Individual hormones were analyzed with a GLMM including center; combinations with ordinary logistic regression without it. Combinations cluster in protocol centers (5 give levothyroxine to all CBI patients, 4 give methylprednisolone to all), which likely also have stronger donation cultures. Consequences: hospital culture is credited to the drugs, and confidence intervals are too narrow because patients from the same hospital are treated as independent.

### 5d. Only 'winners' displayed

A combination appeared only if it met a minimum N, FDR-corrected $p < 0.05$ and a CI ratio < 10. Six hormones allow 15 pairs and 20 triples (35 combinations); 8 are shown. A null or harmful combination can never appear, so the figure can only show benefit. (Full results are said to be in Supplemental Table 4; the main figure is still selected on significance.)

### 5e. Inconsistent inclusion threshold

**[Error]** Methods: $N \geq 75$, justified by a power calculation, with the claim that all 8 combinations were adequately powered. Figure 2 footnote: $N \geq 52$. Three displayed combinations are below 75: methylprednisolone + DDAVP (61), DDAVP + levothyroxine (56), methylprednisolone + vasopressin + DDAVP (52).

## Limitation 6: insulin behaves oddly in the model

Insulin was given to 27.9% of donors and 24.7% of non-donors ($p = 0.353$), yet its adjusted OR is 3.33 (1.18 to 9.37). Possible, but unexplained, and it shows how much estimates depend on modeling choices. Insulin marks hyperglycemia, a different clinical state, rather than a hormone strategy.

## Limitation 7: missing data are not random

Of 1,071 patients with a known outcome, 272 were dropped for any missing covariate, leaving 799 (25% lost) from 29 of 38 centers. Missingness concentrates in the no-HRT group (mechanism missing 15 vs 1; age 7 vs 3; death location unknown 35 vs 6), so the analyzed sample differs from the full population.

## Limitation 8: mediation analysis rests on weak assumptions

Reported: about one third of the effect via progression to brain death (ACME 0.061), two thirds direct (ADE 0.097). But brain death is practically a prerequisite for DBD; mediation assumes no unmeasured mediator-outcome confounding (hemodynamics, OPO involvement, family approach affect both); and splitting a biased total effect just divides a biased number.

## Limitation 9: limited outcome

Donation yes/no only. No organs per donor, organs transplanted, graft function or recipient survival. A drug could increase donors while lowering organ quality, and this design could not detect it.

## Limitation 10: conclusions exceed the data

A routine-use recommendation from observational data affected by the biases above. The NEJM 2023 levothyroxine RCT in brain-dead donors did not increase hearts transplanted. The equity interpretation of demographic differences is not supported by direct data.

## Other errors (annotated in the article PDF)

- **[Error]** Called a 'trial' and 'prospective' in Methods but 'retrospective' in Limitations.
- **[Error]** Table 3: circulatory deaths 283 + 319 = 602, not 646; denominators 685 + 411 = 1,096, not 1,140.
- **[Error]** Hydrocortisone $p < 0.001$ in text vs 0.023 in Figure 2 (the CI 1.23 to 15.78 fits 0.023).
- **[Error]** DDAVP + levothyroxine CI printed as 0.55 to 20.71; Figure 2 shows 4.55.
- **[Error]** Conclusion lists hydrocortisone (not in the HRT definition) and omits insulin.
- **[Method]** Post hoc power analysis while citing Hoenig and Heisey, who argue against it; 'no interaction, so additive' from an underpowered test.

## Summary: which way each limitation pushes

| Limitation | Effect on result | Severity |
|---|---|---|
| Confounding by donation intent | Inflates HRT effect | High |
| Immortal time bias | Inflates HRT effect | High |
| Unknown timing (reverse causation) | Inflates, especially methylprednisolone | High |
| OR read as RR | Effect looks larger | Moderate |
| Combination analysis | Ranking unsupported | Moderate |
| Non-random missing data | Unpredictable, likely inflates | Moderate |
| Mediation assumptions | 'Direct effect' unreliable | Moderate |
| Binary outcome | Organ quality not assessed | Moderate |

***How to say it:*** *Almost every bias points the same way, toward making HRT look better, so the true effect is probably much smaller and could be null. The study defines exposure as any hormone at any time, so part of the association may be reverse causation. It justifies a target trial emulation with time-varying exposure and documented timing relative to brain death, or an RCT, not a change in practice.*

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# Spinoplastic surgical intervention for the treatment of complex spinal disorders and cancer: An initial cohort of 30 vascularized bone grafts

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**Summary** *Introduction:* Complex spinal reconstruction carries high risks for wound and fusion complications. Spinoplastic techniques using vascularized bone grafts may expand the plastic surgeon's role beyond soft-tissue closure. In this study we report outcomes from an initial cohort.
*Methods:* Single-center retrospective chart review of spinal reconstructions between October 2022 and November 2024 was conducted. Primary outcomes included patient symptoms, wound complications, drain output, operative time, and need for revision surgery. Descriptive statistics were utilized to summarize patient outcomes. Descriptive statistics and univariate comparisons were performed against a non-vascularized bone graft control cohort.
*Results:* A total of 30 vascularized bone grafts were performed in 18 patients. A control cohort of 18 patients was identified for comparison. No statistically significant differences were observed between vascularized bone graft and non-vascularized bone graft cohorts in terms of length of hospital stay ($p = 1.00$), operative time ($p = 0.204$), or estimated blood loss

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(p = 0.330). At the time of most recent follow-up, 14 patients (77.8%) reported substantial improvement with regards to pain, three reported stable symptoms, and one patient experienced worsening of symptoms at the most recent follow-up visit.
*Conclusion:* The use of VBGs was associated with a low complication rate ~~and high rates of~~ subjective patient functional improvement. Furthermore, their use does not lead to significantly increased operating room time, intraoperative blood loss, or hospitalization length when compared to a control cohort undergoing similar surgeries without their use. Vascularized bone grafts appear to be a safe and useful adjunct to augment complex spinal surgeries and promote high-quality patient outcomes.
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Patients undergoing complex spinal reconstructive surgery encounter significant challenges; as prior studies have shown that greater than 16% of this patient population can suffer from wound and healing complications.[1,2] The risk of pseudoarthrosis (failure of bony fusion) is a surgical complication of spinal fusion surgery, and rates can exceed 35% depending on the levels of spinal fusion and other patient-specific factors.[3] Patients at an elevated risk of spinal surgical complications include individuals with a history of prior spine surgery, infection, multiple spinal levels involved, or comorbidities such as need for immunosuppressants, smoking history, osteoporosis, diabetes, malnutrition, and obesity.[3,4]

Traditionally, plastic surgeon involvement in spinal surgery has focused on optimization of post-surgical wound healing in complex spinal cases, such as the use of transposition flaps to assist in coverage of a large spinal wound.[5,6] Spinoplastic surgery, however, presents an opportunity for plastic surgeons to play a more active role in the spinal reconstructive process. The term "Spinoplastic surgery," as detailed in this manuscript, involves the use of vascularized bone grafts (VBGs), segments of bone which are transferred on a pedicled muscular attachment and perfused by their associated microvasculature.[7,8] The bone graft is then fixated at the level of spinal reconstruction for enhanced arthrodesis. VBGs offer significant advantages over commonly used non-vascularized bone grafts by directly supplying blood and nutrients to the healing wound bed, thereby reducing healing time and decreasing the risk of hardware failure secondary to infection.[3,5] Maintained vascular supply improves host response to any antibiotic delivery, and improves wound healing. Additionally, unlike standard autografts or allografts, VBGs maintain a living osteogenic cell population with intact vascular networks accelerating graft incorporation and fusion.[3,5] Because of retained viability, VBGs undergo less resorption than their non-vascularized counterparts, preserving long-term mechanical stability critical for extensive complicated reconstructions. Furthermore, prior work demonstrates the ability of Spinoplastic techniques to provide reconstructive support at all levels of the spine through the use of bone harvested from regions including the iliac crest, occiput, rib, and scapula.[7,9-12]

This study aims to expand upon the current body of data pertaining to Spinoplastic techniques by characterizing the post-surgical outcomes of a larger cohort of patients treated using Spinoplastic surgical techniques at our institution. Additionally, we will assemble a control cohort of similar patients who underwent spinal reconstruction without the use of VBGs, thereby comparing several outcomes between VBG and non-VBG cases. The goal of this study will be to present data on an initial cohort of patients treated with vascularized bone grafts, and to better understand how this field can be leveraged to improve outcomes in patients undergoing complex spinal surgery.

## Methods

This study utilized a retrospective chart review to identify spinal reconstructive surgeries which were performed by a single neurosurgeon (M.K.) in conjunction with a single plastic surgeon (E.R.) between October 2022 and November 2024. All surgeries were performed within a single institution. Approval was obtained from the institutional review board (IRB) prior to data collection and the study was performed in accordance with STROBE guidelines to the extent the data and cohort allowed. An electronic medical record query was performed to select for patients who underwent spine surgery augmented by VBGs. VBG was selected for patients deemed at high risk for pseudoarthrosis or compromised healing environments, including those with prior radiation, infection, revision surgery, oncologic resections, or severe multilevel degenerative disease. Current Procedural Terminology (CPT) code 20965, 'harvest and application of vascularized bone graft to spine', was used to identify the patient population and extract data pertaining to their peri-operative and post-operative periods. A cohort of control patients who had undergone spinal surgeries of a similar level of complexity without the use of VBG was also identified. These procedures were performed by the same neurosurgeon during equivalent timeframe.

Patient demographic characteristics including age, sex, basal metabolic index ~~(BMI), smoking status, and history of~~ spine surgery were obtained. Common medical comorbidities such as hypertension, diabetes, and cardiovascular disease were further noted for analysis. Primary outcomes included patient post-operative symptoms (determined at most recent follow-up), wound complications (e.g. infection, hematoma, dehiscence), drain output, surgical time, and need for revision surgery. Additionally, length of

hospital stay, time to postoperative ambulation, subjective symptom assessment, need for reoperation, readmission rate, and average follow-up time was collected across both cohorts. All VBG patient charts were reviewed for subjective symptoms (e.g. improvements in back pain), however, functional outcomes were not compared between cohorts.

A cohort of 18 control patients was identified in order to match the size of the intervention cohort. These were patients who underwent similarly complex spinal fusion surgery without the use of VBGs, defined as fusions involving five or more spinal levels and closed with transposition muscle flaps by the Plastic Surgery team. The control patients resembled the VBG cohort in that the complexity of their operations would have made them suitable candidates for VBG utilization, had they also presented with the comorbidities warranting a spinoplastic approach.

Descriptive statistics were utilized to summarize patient outcomes. An adequately sized control cohort (consisting of non-VBG patients) was identified to power analyses. Univariate analysis was performed in SPSS to compare between VBG and non-VBG cohorts, using Mann-Whitney *U* tests to assess continuous variables and Fisher's Exact Test to evaluate categorical variables. P-values were determined to be significant if they were found to be less than or equal to 0.05.

## Results

### Primary outcomes

A total of 30 VBGs were performed in 18 patients, of which 25 grafts (83.3%) were harvested from the iliac crest, three from the ribs, and two from the occipital bone (Table 1). The cohort had a mean age of 63.0 +/- 15.8 years, and 12 patients (67%) had a history of previous spinal surgery. All patients underwent at least one muscle transposition flap for spinal closure in addition to the VBG. Bone Morphogenic Protein (BMP) was utilized for stimulation of osteogenesis in six (35%) patients. Table 2 details the various types of VBGs which have been used by our institution in the past for spinal reconstruction. Furthermore, **Video 1** demonstrates the general technique for the harvest and inset of an iliac crest VBG, the most commonly deployed technique at our institution.

Supplementary material related to this article can be found online at doi:10.1016/j.bjps.2026.03.004.

At the time of most recent follow-up, 14 patients (77.8%) reported subjective improvement with regards to pain, three reported stable symptoms, and one patient experienced worsening of symptoms at the most recent follow-up visit. Notably, several patients were documented as experiencing significant improvements to function and pain in follow-up clinic documentation, including a patient who was "pain-free for the first time in 16 years" and another who experienced improvements "with regard to not only pain but also activity." One patient reported being able "to stand upright" for the first time since their initial surgery, displaying improved postural stability and mobility. Most patients denied new neurological effects, with the exception of one patient who reported right foot drop following surgery. First ambulatory event occurred at a mean 3.2 days post-operatively, and average length of hospital stay post-reconstruction was 6.4 days.

Of note, the control cohort had a mean age of 62 +/- 14.1 years. Subjective functional improvement was noted in 10 (56%) patients from the control cohort at most recent follow-up. Given that functional outcomes were derived from post-operative follow up notes, no attempt at statistical comparison was made between these data.

Upon univariate analysis, no statistically significant differences were observed between VBG and non-VBG cohorts in terms of length of hospital stay (p = 1.00), operative time (p = 0204), estimated blood loss (p = 0.330), or post-operative days until ambulation (p = 0.243) (Figure 1) (Table 3). One patient remained intubated for an extended period of time after surgery and was not able to ambulate until 70 days postoperatively, and another patient was unable to ambulate due to limb malformation secondary to

**Table 1** Vascularized bone graft patient demographics.

| Vascularized bone graft patients surgical date ranges: 10/2022-11/2024 | | |
|---|---|---|
| N Patients = 18 | Average | SD |
| **N Vascularized Bone Grafts = 30** | | |
| **Cohort Characteristics** | | |
| Age (years) | 63.0 | 15.8 |
| Male/Female | 11/7 | - |
| BMI (kg/m$^2$) | 31.8 | 6.4 |
| Hypertension | 11 (61.1%) | - |
| Diabetes | 2 (11.1%) | - |
| Smoking History | 5 (27.8%) | - |
| History of Cardiovascular Disease | 11 (61.1%) | - |
| History of spinal surgery | 12 (66.7%) | - |
| Other major comorbidity | 10 (55.6%) | - |
| **Surgical Characteristics** | | |
| Length of Surgery | 8 h 38 min | 2 h 26 min |
| Muscle Transposition Flap | 16 (100%) | - |
| BMP Used | 6 (33.3%) | - |
| Morselized Bone Graft Used | 16 (88.9%) | - |
| Unilateral VBG | 8 (44.4%) | - |
| Bilateral VBG | 10 (55.6%) | - |
| Total Estimated Blood Loss | 1636 mL | 1593 mL |
| Number of Post-Surgical Drains | 2.7 | 0.9 |
| **Post-Operative Course** | | |
| Length of Hospital Stay (days) | 7.5 | 5.4 |
| Average Days Until Post-Operative Ambulation | 3.2 | 1.9 |
| Symptoms Improved at Most Recent Visit | 14 (77.8%) | - |
| Average Plastic Surgical Follow-Up Length (months) | 1.9 | 1.9 |
| Average Neurosurgical Follow-Up Length (months) | 13.0 | 7.1 |
| Reoperation by Plastic Surgery | 4 (22%) | - |
| Reoperation by Neurosurgery | 8 (44.4%) | |
| Any Readmission | 9 (50%) | - |

**Table 2** Comparative summary of vascularized bone grafts and spinal reconstruction levels.

| VBG Type | Blood Supply | Surgical Complexity | Appropriate Clinical Scenario |
|---|---|---|---|
| Iliac Crest VBG (IC-VBG) | Quadratus lumborum | Moderate | Posterolateral spinal reconstruction from **T12-S1** |
| Scapular VBG (S-VBG) | Transverse cervical artery | Moderate-high | Posterior cervical-thoracic reconstruction from the **occiput to T8** |
| Rib VBG (R-VBG) | Intercostal vessels | Moderate | Posterior spinal defects with rib curvature alignment* |
| Occipital VBG (O-VBG) | Deep cervical artery | High | Posterior cervical reconstruction from **C1 to T1** |
| Spinous Process (SP-VBG) | Paraspinal muscles | Moderate | Posterolateral fusion augmentation, can be utilized from the **occiput to T8** |
| Clavicular VBG (C-VBG) | Occipital & superior thyroid artery | Very high | Anterior cervicothoracic reconstruction from **C2 to T2** |

* Specific spinal levels depend on which rib has been selected for harvest.

V9

~~tetra-amelia syndrome.~~ These data points were excluded as outliers for the purposes of statistical analysis.

## Intra-operative characteristics

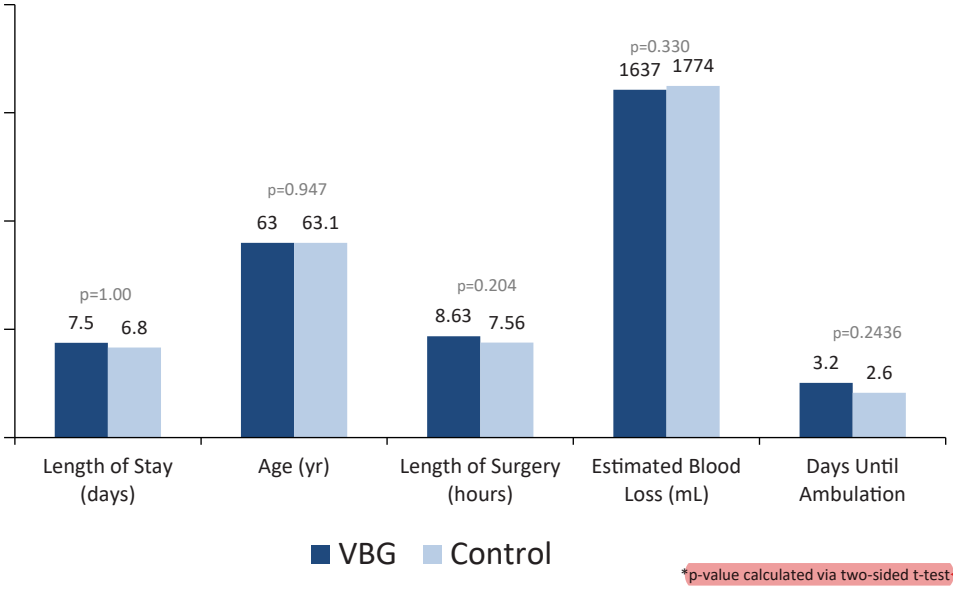
Size of VBG harvested was documented in the operative notes of eight of the 30 (27%) total VBGs performed. Average VBG size was 4.75 cm, with the smallest VBG performed being 3 cm and the largest being 10 cm. Unilateral placement of VBG occurred in eight (44%) patients, compared with bilateral placement in the remaining patients (56%). One patient underwent bilateral graft placement at two separate spinal levels, receiving rib-based VBGs at T7-T8 and iliac crest-based VBGs at L5-S1. Transposition flaps were utilized for spinal wound closure in all examples, often including latissimus dorsi, paraspinous muscle, or trapezius flaps (or a combination of these techniques) to ensure adequate soft tissue coverage.

Nine patients had estimated blood loss (EBL) related to their VBG placement documented in their Plastic Surgery operative report, with an average blood loss of 255.5 mL ± 150.9 mL. Average total case blood loss was 1636.7 mL ± 1593.8 mL. Intraoperative blood products were required for eleven (61%) patients, with a range of one unit packed red blood cells (PRBCs) to 14 units PRBCs.

## Post-operative course

Twelve patients (67%) required postoperative blood transfusions. Operative drains were placed for all patients, with a minimum of one drain to a maximum of four drains placed, mean of 2.7 drains. The average length of time to final drain removal was 23.6 days post-operatively. Post-operative drain output was recorded daily for up to 13 days. On day 1 the average output across all drains was 245.84 mL ± 68.9 mL, and output gradually decreased over subsequent days. One case of wound infection with dehiscence requiring hardware removal and tissue debridement was noted in this study. No other cases of wound infection or dehiscence were reported, nor were there any reported episodes of hematoma formation. The cohort's average time from surgery to most recent follow-up was 10.9 +/- 5.0 months.

Eight patients ultimately underwent revision surgery, four of which included plastic surgical interventions. Of




V10

**Figure 1** Graphical representation of the Vascularized Bone Graft (VBG) Cohort compared to the control (non-VBG) cohort in terms of pertinent outcomes. Notably, there was no statistically significant difference identified between these characteristics.

**Table 3** Comparison of vascularized bone graft patient cohort to control cohort.

| | VBG cohort mean (n = 18) | SD | Control cohort mean (n = 18) | SD | p-value* |
|---|---|---|---|---|---|
| **Length of Stay (days)** | 7.50 | 5.36 | 6.76 | 3.17 | 1.00 |
| **Age (years)** | 63.0 | 15.8 | 63.1 | 14.4 | 0.947 |
| **BMI (kg/m$^2$)** | 31.8 | 6.4 | 27.3 | 5.8 | 0.044 |
| **Length of Surgery (h)** | 8 h 38 min | 2 h 26 min | 7 h 34 min | 1 h 46 min | 0.204 |
| **Estimated Blood Loss (mL)** | 1637 mL | 1594 mL | 1774 mL | 1358 mL | 0.330 |
| **POD Until Ambulation** | 3.2 days | 1.9 days | 2.6 days | 1.9 days | 0.243 |

Abbreviations: POD, Post-operative days; SD, Standard deviation; VBG, vascularized bone graft.
* P-values calculated using the Mann-Whitney *U* test due to non-normally distributed datasets

V11

note, there was no statistically significant re-operation rate identified among the intervention and control cohorts on Fisher's Exact Test analysis (p = 0.740). One VBG patient, for example, underwent revision surgery for removal of loosened screws four months after initial surgery. During repeat operation, it was noted that the patient had attained bony fusion of bilateral iliac crest VBGs at the transverse processes of L5 vertebrae. Notably, another patient who ultimately underwent further neurosurgical intervention for loose hardware nearly two years after initial VBG surgery underwent pre-operative imaging, and in this study there was noted to be persistent bony fusion of the graft at the site of fixation (Figure 2). It is important to note that bony fusion was not an explicit outcome of this study: post-operative imaging is not recommended as standard of care for these patients, therefore any bony fusion was identified as an "incidental finding" on imaging performed for a separate reason. Lastly, donor site morbidity was not systematically documented in the retrospective record and therefore was not analyzed.

V12

V13

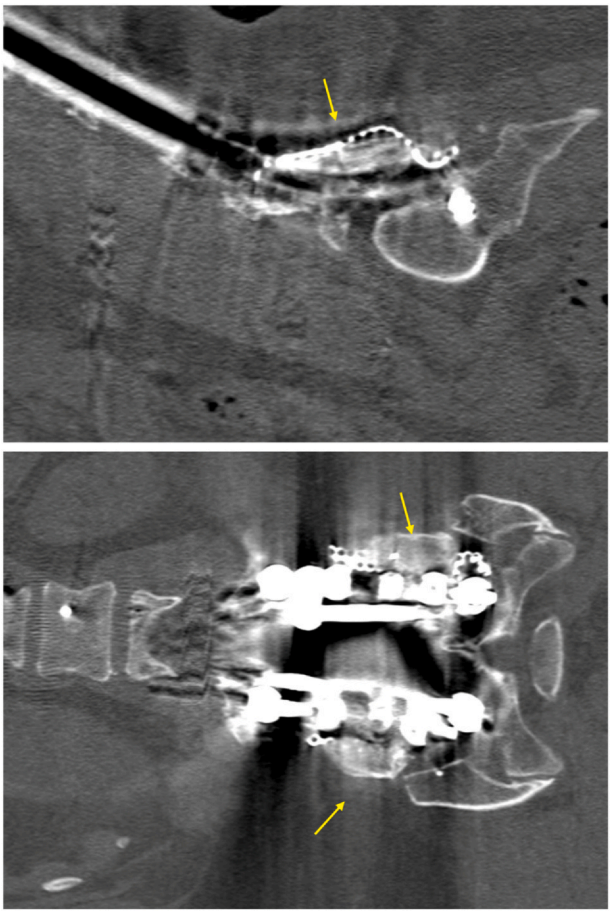


**Figure 2** Coronal and sagittal computed tomography (CT) scan views of a patient who underwent vascularized bone graft nearly two years prior to the scan, demonstrating bony fusion at the level of fixation with healthy-appearing cortex and medullary iliac crest bone graft.

## Discussion

This collection of 30 VBGs serves as the first case series documenting the use of Spinoplastic surgical techniques in patients undergoing complex spinal reconstruction. This cohort demonstrated encouraging results in terms of long-term pain and functional outcomes, with the vast majority (nearly 80%) of patients reporting subjective improvements at their most recent follow-up appointment. These patients are typically followed by the Plastic Surgery service for wound care and drain management, but long-term follow-up is performed by the neurosurgical team for bony arthrodesis and neurological outcomes. In this cohort of patients, the mean neurosurgical follow-up length at time of writing was approximately 13 months. However, though results presented here are exciting, further studies are mandatory in order to demonstrate continued long-term functional improvement, as well as radiographic confirmation of bony fusion and arthrodesis. Of note, 'functional outcomes' were derived from the descriptive text in post-operative progress notes, and therefore this was not a standardized metric in this study. As such, the results should not be interpreted as evidence of comparative efficacy.

V14

V15

In terms of post-operative management, Spinoplastic surgical patients were monitored inpatient for an average length of stay of 7.5 days. Throughout this timeframe patients are encouraged to ambulate when possible (our cohort first ambulated with physical therapists after a mean 3.2 days) in order to promote a favorable biomechanical healing environment. Surgical drains are removed after two consecutive days of less than 30ccs of serosanguinous output. The majority of patients were discharged with either one or two drains in place, which were subsequently removed without complication in clinic once meeting criteria for removal. We observed very few wound complications in this cohort, likely due to meticulous spinal closure

using a combination of transposition flaps and incisional wound VACs.[6]

The incorporation of Spinoplastic surgery in the context of challenging spinal cases may be a valuable way to improve bony fusion in patients. Scenarios including radiated wound beds, oncologic resections, poor bone stock, and immunosuppressed patients present particularly challenging cases and represent scenarios when the authors consider implementing Spinoplastic techniques.[13-18] These techniques were also utilized in several patients undergoing primary spinal reconstruction, often in the context of severe degenerative disease or oncologic resection. The goal of using prophylactic VBGs in many of these patients is the historically high rate of pseudoarthrosis in long-segment degenerative fusions or in spinal oncology patients who may require large-segment bone resections or those requiring postoperative radiation.

One example is a 60-year-old patient who presented with a lytic cervical mass: CT imaging revealed involvement of right paraspinal muscles, lamina, spinous process, and moderate-severe canal stenosis without cord compression, which ultimately required C2 corpectomy with decompression and fusion from the occiput to C5. In this case, occipital VBGs were chosen as an adjunct to aid with fusion. This patient ultimately went on to experience satisfactory healing without complication and noted an overall improvement in his subjective symptoms compared with before surgery. Spinal oncologic resections have been documented to possess general complication rates hovering near 50%, therefore any ability to lessen this risk such as using VBGs warrants further exploration.[19-21]

Adding to the utility of VBGs is their ability to be customized to the specific anatomic needs. Prior studies have demonstrated the ability of vascularized bone grafts to be procured for all levels of the spine, suggesting that this technique can be used in spinal reconstructive circumstances spanning the entirety of the spinal column.[22-26] The introduction of a vascularized bone graft into the wound bed offers well-perfused tissue to augment a potentially tenuous arthrodesis environment, and in this cohort we observed strong outcomes in both primary and revision cases. It is important to note that free vascularized bone flaps (e.g. free fibular flaps) still play an important role in certain spinal reconstruction, particularly in scenarios which require a more bulky or load-bearing or reconstruction.

Of note, given that the standard of care post-operative imaging for these patients (as guided by our neurosurgery colleagues) typically does not include obtaining CT or MRI scans, we generally do not have radiologic evidence of VBG bony fusion among these patients. However, Figure 2 displays the imaging of a patient who underwent an unrelated neurosurgical procedure nearly two years post-VBG, in which a CT scan clearly demonstrates bony fusion of the joint where the VBG was placed in the index operation. Spinal X-Rays were obtained in the majority of patients which revealed signs of VBG bony cortex at several months post-operatively, however, artifact due to hardware makes definitive assessment difficult with this modality. Future studies may incorporate the use of imaging to monitor post-operative bony integration to allow for further direct comparison of arthrodesis between techniques.

There are several limitations to our study, first of which is the retrospective nature of this chart review and the inherent challenges in drawing conclusions from a study of this type. The patient cohort also represents a medically and surgically diverse group of individuals, including patients who received radiation to the affected areas, subsequent infections, and hardware removal. This heterogeneity may present confounding factors to our assessment of post-operative outcomes, including assessment of the subjective symptoms of general pain and mobility.

Finally, the relative lack of radiographic evidence of VBG bony fusion requires us to judge the outcome of successful bony fusion largely on clinical symptoms (e.g., lack of pain, functionality), and future studies will be designed to incorporate follow-up imaging to assess the long-term bony fusion rates of VBGs. Due to the lack of a systematic, standardized postoperative imaging protocol, as well as limited time to follow-up, this study is unable to establish evidence for fusion rates or time to fusion of the VBGs. Despite these limitations, the authors are confident that these data highlight the utility of VBGs as a new rung to the reconstructive ladder when approaching complex spinal reconstruction.

## Conclusion

In conclusion, pedicled VBG use in complex spinal reconstruction appears feasible and is not associated with increased operative time, blood loss, or length of stay relative to a control cohort of similarly complex non-VBG patients. The findings support the use of VBGs as a potentially useful adjunct in selected cases, however, further comparative prospective studies with standardized clinical and radiographic follow-up are required for evidence of comparative efficacy. Future research will be essential to further refine the appropriate surgical applications of this technique, and to assess its long-term outcomes in this population.

## Ethical approval

Deemed 'exempt' via Mayo Clinic Institutional Review Board, IRB study #24-008578.

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None.

## Declaration of Competing Interest

Maziyar Kalani, MD include: Medtronic (Consulting), Globus (Royalties, Consulting), Surgical Theater (Consulting), and Carlsmed (Consulting). No relevant conflicts of interest pertaining to this study. W. Nicholas Jungbauer, MD, Daniel Vernik, MS, Dylon Buchanan, MS, Krishna Unadkat, MS, Jonathan L. Jeger, MB BCh, Nicole Van Spronsen, MD, Jenna

Meyer, MD, Michael Bohl MD, Sebastian Winocour, MD, MBA, Alexander E. Ropper, MD, and Edward Reece, MD MBA have no financial interests or conflicts of interest to disclose.

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# 4. Spinoplastic surgery: vascularized bone grafts (JPRAS 2026)

## Background

- **Problem:** complex spinal fusion (multilevel, revision, irradiated, oncologic) has high wound complication rates and pseudarthrosis (failed fusion) that can exceed 35%.
- **Non-vascularized graft** (autograft/allograft): dead scaffold, heals slowly by creeping substitution and resorbs. **Pedicled VBG:** bone rotated on a muscle that keeps its blood supply, so it stays alive and heals like a fracture. Most common: iliac crest on the quadratus lumborum; alternatives rib, occiput, scapula, spinous process, clavicle.
- **'Spinoplastic surgery'** was coined by the Reece/Ropper/Winocour group (Baylor, now Mayo Arizona); most cited literature, including cadaveric and technique papers, comes from that group.
- **Terms:** BMP (osteoinductive adjunct); transposition muscle flaps for closure; free fibula flap for load-bearing defects.

## What it reports

30 VBGs in 18 patients (25 iliac crest, 3 rib, 2 occipital) vs 18 controls with complex fusion (5 or more levels) closed with muscle flaps. Same neurosurgeon and plastic surgeon, single center, October 2022 to November 2024. No significant difference in length of stay (p = 1.00), operative time (p = 0.204), blood loss (p = 0.330) or days to ambulation (p = 0.243); 14 of 18 (77.8%) reported subjective improvement.

The paper makes two claims, and every limitation below weakens one or both: **claim 1 (safety)**, VBG adds no operating time, blood loss or length of stay; **claim 2 (benefit)**, VBG gives high rates of improvement.

## Limitation 1: the groups are not comparable (confounding by indication)

VBG was chosen for the patients the surgeons judged at highest risk. Controls were patients who would have been candidates 'had they also presented with the comorbidities warranting a spinoplastic approach', that is, patients defined by lacking the high-risk features.

| | VBG group | Control group |
|---|---|---|
| Selection rule | High risk: prior radiation, infection, revision, oncologic resection, severe multilevel disease | 5 or more levels, closed with muscle flaps, without the high-risk features |
| How chosen | CPT code 20965 | Not stated (number picked 'to match the size') |
| Matching | None | None |

Only age (63.0 vs 63.1) and BMI (31.8 vs 27.3, **p = 0.044**) are compared. Diabetes, smoking, prior spine surgery, radiation, cancer and number of levels are reported for the VBG group only. **Direction: unclear**, and that is the problem: sicker VBG patients bias complications against VBG, while an unstated control selection can bias either way.

## Limitation 2: too small to show 'no difference' (supports claim 1 falsely)

With 18 patients per group only very large differences can be detected, so 'not significant' means 'could not tell', not 'the same'. The authors say the control cohort was sized 'to power analyses', but no power calculation is reported.

| Outcome | VBG | Control | Difference (approx. 95% CI) | Smallest detectable difference* |
|---|---|---|---|---|
| Operative time | 8 h 38 min | 7 h 34 min | +64 min (-22 min to +2.5 h) | about 2 h |
| Blood loss | 1,637 mL | 1,774 mL | -137 mL (-1,100 to +830 mL) | about 1,400 mL |

| Outcome | VBG | Control | Difference (approx. 95% CI) | Smallest detectable difference* |
|---|---|---|---|---|
| Length of stay | 7.5 days | 6.8 days | +0.7 days (-2.1 to +3.6) | about 4 days |

*80% power, alpha 0.05, using the paper's SDs. CIs and detectable differences are approximations calculated from the reported means and SDs, not values reported by the authors. The data are compatible with VBG adding up to 2.5 hours of surgery or 3.6 days in hospital. **Direction: toward 'no difference', which is exactly what claim 1 needs.**

## Limitation 3: fusion, the reason to use VBG, was never measured (undermines claim 2)

VBG exists to prevent pseudarthrosis, which the authors cite as exceeding 35%. The paper states that 'bony fusion was not an explicit outcome': no routine CT, and X-rays limited by hardware artifact. The only fusion evidence comes from two patients, one seen at revision for **loosened screws** at 4 months and one on CT before reoperation for **loose hardware** at about 2 years. Both had hardware failure, and two cases cannot establish a fusion rate. Analogy: a study of a new heart valve that reports operating time but never checks whether the valve works.

## Limitation 4: 'improvement' is anecdotal (inflates claim 2)

- No pain score (VAS/NRS), no disability index (ODI), no quality-of-life measure: improvement was judged from clinic notes.
- Not blinded: the notes were written by the treating surgeons.
- Controls assessed the same way improved in 10 of 18 (56%); no statistical comparison was made, yet the abstract highlights 77.8%.
- Quotes such as 'pain-free for the first time in 16 years' are selected anecdotes.
- The authors write that results 'should not be interpreted as evidence of comparative efficacy', but the abstract concludes VBGs 'promote high-quality patient outcomes'.

## Limitation 5: 'low complication rate' vs the actual numbers

| Event (VBG patients, n = 18) | n (%) |
|---|---|
| Any readmission | 9 (50%) |
| Reoperation by neurosurgery | 8 (44.4%) |
| Reoperation including plastic surgery | 4 (22%) |
| Intraoperative transfusion | 11 (61%), up to 14 units |
| Postoperative transfusion | 12 (67%) |
| Wound infection with hardware removal | 1 |
| Hardware loosening | at least 2 |
| New foot drop | 1 |
| Prolonged intubation, not ambulating until day 70 | 1 |
| Time to final drain removal | 23.6 days (mean) |

The introduction cites a 16% wound complication rate for this population. The reoperation comparison (p = 0.740) is reported without the control group's numbers, and control readmissions are not reported at all.

## Limitation 6: selective outcome reporting

| Primary outcome promised in Methods | Compared with controls? |
|---|---|
| Symptoms | No ('no attempt at statistical comparison') |
| Wound complications | No |
| Drain output | No (VBG only) |
| Operative time | Yes |
| Revision surgery | p value only, no control numbers |

Length of stay, blood loss and ambulation, which were not listed as primary outcomes, were compared instead.

## Limitation 7: follow-up too short for fusion failure

Plastic surgery follow-up 1.9 months; overall 10.9 ± 5.0 months (Results); neurosurgery 13.0 months (Table 1). Pseudarthrosis usually becomes evident at 12 to 24 months, so failures may not have shown up yet. Results (10.9) and Discussion ('approximately 13') also disagree. **Direction: underestimates failures.**

## Limitation 8: co-interventions make it impossible to isolate VBG

Besides VBG: morselized bone graft 16 (88.9%), BMP 6 (33.3%), muscle transposition flaps in all, and incisional wound VAC (credited by the authors for the few wound complications). Patients were heterogeneous: oncologic, revision, radiated and degenerative cases, from the occiput to the sacrum, unilateral or bilateral grafts.

## Limitation 9: missing and unrecorded data

- VBG size documented for 8 of 30 grafts (27%); VBG-related blood loss for 9 of 18 patients.
- **Donor-site morbidity not recorded**, although iliac crest or rib harvest carries pain, hernia and nerve risks: the most relevant missing safety outcome.
- Two patients excluded from the ambulation analysis as outliers, decided after seeing the data.

## Limitation 10: single team, single center, possible conflicts

All operations by one neurosurgeon and one plastic surgeon, from the group that coined 'spinoplastic surgery': results may reflect their skill and learning curve. The neurosurgeon reports consulting for Medtronic and Globus (spinal hardware and biologics); 'no relevant conflicts' is declared.

## Internal errors (annotated in the article)

- **[Error]** Length of stay 6.4 days in text vs 7.5 in Tables 1 and 3.
- **[Error]** BMP 'six (35%)', but 6/18 = 33.3%.
- **[Error]** Muscle flaps '16 (100%)' in Table 1, but 16/18 = 89%, and the text says all patients had flaps.
- **[Error]** Figure 1 says t-test; Table 3 says Mann-Whitney U.
- **[Error]** 'p = 0204' (0.204); 'basal metabolic index' (body mass index); references 10 and 26 duplicated.
- **[Method]** Claims to be 'the first case series', although the same group has published earlier reports.

## Summary: what each limitation does to the two claims

| Limitation | Effect | Severity |
|---|---|---|
| Non-comparable groups | Comparison uninterpretable | High |
| Underpowered | Makes 'no difference' (claim 1) look stronger | High |

| Limitation | Effect | Severity |
|---|---|---|
| Fusion not measured | Claim 2 untested | High |
| Subjective, unblinded outcomes | Inflates claim 2 | High |
| Complications understated | Safety overstated | Moderate |
| Selective outcome reporting | Emphasis on favorable outcomes | Moderate |
| Short follow-up | Underestimates failures | Moderate |
| Co-interventions | Effect not attributable to VBG | Moderate |
| Missing donor-site data | Key harm unknown | Moderate |

***How to say it:*** *This is a feasibility report, level IV evidence. The safety claim rests on an underpowered comparison with non-comparable controls; the data are compatible with up to 2.5 hours of extra operating time. The benefit claim rests on unblinded clinic notes, while fusion, the outcome VBG is meant to improve, was never measured. Half the patients were readmitted and 44% needed reoperation. What is needed is a prospective comparative study with matched controls, a prespecified sample size, CT-confirmed fusion at 12 and 24 months, validated outcome scores and donor-site morbidity.*

# Glossary

| Term | Meaning |
|---|---|
| Confounding by indication | The reason a treatment is given is itself linked to the outcome (e.g. HRT given to likely donors) |
| Immortal time bias | Exposed patients must survive long enough to be exposed, guaranteeing them a survival period the unexposed do not get |
| Reverse causation | The outcome (or its pathway) causes the exposure rather than the other way round |
| Landmark analysis | Analysis starting at a fixed time point, classifying exposure up to then, to remove immortal time |
| Target trial emulation | Designing an observational analysis to mimic a specified randomized trial (clear eligibility, time zero, exposure) |
| OR vs RR | With common outcomes, the odds ratio exaggerates the relative risk |
| Intention to treat | Analyze patients in the groups they were randomized to, without post-randomization exclusions |
| MCID | Minimal clinically important difference; smallest change patients perceive as meaningful |
| MME | Morphine milligram equivalents; standardizes opioid doses (Norco 5/325 = 5 MME) |
| TSA / RIS | Trial sequential analysis / required information size: whether cumulative evidence is conclusive |
| GRADE | Framework rating certainty of evidence (high, moderate, low, very low) |
| AMSTAR 2 | Tool for appraising the quality of systematic reviews |
| GLMM, random intercept, ICC | Mixed model allowing each center its own baseline; ICC = share of variance due to center |
| ACME / ADE | Average causal mediation effect (indirect) / average direct effect |
| OPO | Organ Procurement Organization: evaluates donors, approaches families, manages donors, allocates organs |
| DBD / DCD | Donation after brain death / after circulatory death |
| VBG | Vascularized bone graft: bone transferred with its blood supply |
| Pseudarthrosis | Failure of bony fusion after spinal arthrodesis |